DEVELOPMENT OF AN AI-BASED PREDICTIVE MODEL FOR THE DETECTION AND DIAGNOSIS OF BEARING FAULTS

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ABSTRACT

Rolling element bearings are critical components in rotating machinery, and their failure can result in costly equipment downtime, reduced productivity and increased maintenance expenses. Traditional maintenance strategies often detect faults only after significant damage has occurred, highlighting the need for intelligent fault diagnosis systems capable of supporting predictive maintenance. This study investigated the application of supervised machine learning techniques for bearing fault diagnosis using statistical vibration features extracted from the Case Western Reserve University (CWRU) bearing dataset. The study followed a structured machine learning workflow comprising exploratory data analysis, data preprocessing, feature importance analysis, model development, comparative evaluation, hyperparameter optimisation and model deployment. Five supervised machine learning algorithms, namely Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours and Support Vector Machine, were developed and evaluated using a consistent experimental framework. Comparative analysis showed that all evaluated models achieved classification accuracies exceeding 90%, demonstrating the effectiveness of statistical time-domain vibration features for multiclass bearing fault classification. Among the evaluated algorithms, the Random Forest classifier achieved the best overall performance. Following hyperparameter optimisation using GridSearchCV with five-fold cross-validation, the final Random Forest model attained a test accuracy of **94.57%**, together with high precision, recall and F1-score values, indicating excellent predictive capability. The trained model was subsequently saved for future deployment in predictive maintenance applications. The findings demonstrate that supervised machine learning provides an accurate, reliable and computationally efficient approach to vibration-based bearing fault diagnosis. The developed framework contributes to intelligent condition monitoring and has the potential to support early fault detection, improve equipment reliability, reduce maintenance costs and enhance maintenance decision-making in industrial environments.

Keywords: Bearing fault diagnosis, machine learning, predictive maintenance, vibration analysis, Random Forest, statistical feature extraction.

Chapter One

1 INTRODUCTION

1.1 Background of the Study

Rolling element bearings are among the most widely used mechanical components in rotating machinery due to their ability to reduce friction while supporting radial and axial loads. They are essential components in industrial equipment such as electric motors, pumps, compressors, turbines, gearboxes, conveyors, machine tools and various manufacturing systems. The performance, reliability and operational efficiency of these machines depend significantly on the health condition of their bearings. Consequently, bearing failures can result in excessive vibration, reduced machine efficiency, unplanned downtime, and increased maintenance costs and, in severe cases, catastrophic equipment failure.

The increasing dependence on rotating machinery across industries has made equipment reliability a major concern for manufacturers and maintenance engineers. Traditionally, industrial maintenance has been performed using either corrective maintenance, where repairs are carried out only after a failure occurs, or preventive maintenance, where components are serviced or replaced at predetermined intervals regardless of their actual condition. Although preventive maintenance reduces the likelihood of unexpected failures, it often leads to unnecessary maintenance activities and increased operational costs. To overcome these limitations, industries are increasingly adopting predictive maintenance, a maintenance strategy that monitors the condition of machinery and predicts potential failures before they occur, thereby improving equipment availability and reducing maintenance costs.

Among the various condition monitoring techniques available for predictive maintenance, vibration analysis has proven to be one of the most effective methods for detecting defects in rolling element bearings. Bearing faults, including defects on the inner race, outer race and rolling elements, alter the vibration characteristics of rotating machinery. These changes can be captured using vibration sensors and analysed to identify fault conditions at an early stage before significant mechanical damage occurs. Early fault detection not only improves equipment reliability but also enhances operational safety and reduces production losses associated with unexpected machine breakdowns.

Conventional vibration-based fault diagnosis largely depends on manual feature interpretation and the experience of maintenance personnel. While this approach has been successfully applied in many industrial settings, it becomes increasingly challenging when analyzing large volumes of vibration data generated by modern industrial systems. The growing complexity of manufacturing processes and the increasing deployment of intelligent monitoring systems have created the need for automated diagnostic techniques capable of accurately identifying machine faults with minimal human intervention.

Recent advancements in artificial intelligence and machine learning have significantly transformed the field of condition monitoring and fault diagnosis. Machine learning algorithms are capable of learning complex relationships between extracted vibration features and bearing health conditions, enabling automated classification of machine faults with high levels of accuracy. Supervised learning algorithms such as Logistic Regression, Decision Trees, Random Forests, K-Nearest Neighbours and Support Vector Machines have demonstrated considerable success in bearing fault diagnosis because of their ability to recognise hidden patterns within vibration data and generalise effectively to unseen observations.

In this study, machine learning techniques are applied to classify rolling element bearing faults using statistical features extracted from vibration signals obtained from the Case Western Reserve University (CWRU) Bearing Data Center dataset. The extracted features include measures such as mean, standard deviation, root mean square (RMS), skewness, kurtosis and crest factor, which characterize different aspects of the vibration signal. Multiple machine learning algorithms are trained, evaluated and compared using a consistent experimental framework to identify the most suitable classifier for bearing fault diagnosis. The outcomes of this study are expected to contribute to the development of intelligent condition monitoring systems that support predictive maintenance strategies in industrial applications.

1.2 Problem Statement

Conventional maintenance practices, including corrective and preventive maintenance, have several limitations. Corrective maintenance addresses faults only after failure has occurred, often leading to significant economic losses and production interruptions. Preventive maintenance, on the other hand, follows predetermined maintenance schedules without considering the actual condition of the equipment, which may result in unnecessary replacement of healthy components and increased maintenance costs. These limitations have encouraged industries to adopt predictive maintenance strategies that rely on continuous condition monitoring to detect faults before they develop into serious failures.

Vibration analysis has become one of the most widely adopted techniques for bearing condition monitoring because mechanical defects produce measurable changes in vibration signals. However, the accurate interpretation of vibration data remains a challenging task. Traditional diagnostic methods often depend on manual analysis and expert knowledge, making them time-consuming, subjective and less effective when dealing with large volumes of sensor data or complex fault patterns. As industrial systems continue to generate increasing amounts of condition monitoring data, there is a growing need for automated diagnostic approaches capable of providing fast, accurate and reliable fault identification.

Therefore, this study addresses the problem of developing an intelligent bearing fault diagnosis system capable of accurately classifying different bearing fault conditions using statistical features extracted from vibration signals. By evaluating and comparing the performance of multiple supervised machine learning algorithm including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours and Support Vector Machine, this research seeks to identify the most suitable model for supporting reliable and efficient bearing fault diagnosis in predictive maintenance applications.

1.3 Aim of the Study

The aim of this study is to develop and evaluate a machine learning-based bearing fault diagnosis system using statistical features extracted from vibration signals in order to accurately classify rolling element bearing fault conditions and support predictive maintenance in rotating machinery.

1.4 Objectives of the Study

The specific objectives of this study are to:

1. Acquire and prepare the Case Western Reserve University (CWRU) bearing vibration dataset for machine learning analysis.
2. Perform exploratory data analysis to understand the characteristics and distribution of the extracted vibration features.
3. Preprocess the dataset through data cleaning, feature scaling and label encoding to prepare it for model development.
4. Evaluate the importance of the extracted statistical features used for bearing fault classification.
5. Develop and train supervised machine learning models, including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors and Support Vector Machine, for bearing fault diagnosis.
6. Compare the performance of the developed models using appropriate evaluation metrics, including accuracy, precision, recall and F1-score.
7. Optimize the best-performing model through hyperparameter tuning to improve its classification performance.
8. Develop a final bearing fault diagnosis model capable of accurately classifying different bearing fault conditions.

1.5 Research Questions

This study seeks to answer the following research questions:

1. How can vibration signal data be effectively prepared for machine learning-based bearing fault diagnosis?

2. What characteristics are revealed through exploratory analysis of the extracted vibration features?
3. Which statistical vibration features contribute most significantly to bearing fault classification?
4. How do different supervised machine learning algorithms perform in classifying rolling element bearing faults?
5. Which machine learning algorithm provides the best overall performance for vibration-based bearing fault diagnosis?
6. To what extent does hyperparameter tuning improve the performance of the selected machine learning model?

1.6 Scope of the Study

The scope of this study is limited to the development and evaluation of machine learning models for rolling element bearing fault diagnosis using vibration signal data. The study utilizes the Case Western Reserve University (CWRU) Bearing Data Center dataset, which contains vibration signals collected under controlled laboratory operating conditions for healthy bearings and bearings with induced faults.

The investigation focuses on statistical features extracted from the vibration signals, including the mean, minimum, maximum, standard deviation, root mean square (RMS), skewness, kurtosis, crest factor and waveform factor. These features are used as input variables for the development of supervised machine learning models.

Five supervised machine learning algorithms including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors and Support Vector Machine, are developed, trained and evaluated. Their performances are compared using standard classification metrics, including accuracy, precision, recall and F1-score. Hyperparameter tuning is subsequently applied to improve the performance of the best-performing classifier.

This study does not include deep learning techniques, real-time industrial deployment, online condition monitoring or the design of embedded diagnostic hardware. Furthermore, the investigation is limited to the available CWRU dataset and does not consider vibration data collected from industrial environments operating under varying load conditions or different machine configurations.

1.7 Significance of the Study

This study contributes to both the fields of mechanical engineering and intelligent condition monitoring by demonstrating the application of machine learning techniques for bearing fault diagnosis using vibration signal analysis. The findings are expected to provide valuable insights into the effectiveness of different supervised machine learning algorithms for classifying bearing fault conditions.

For the manufacturing and maintenance industries, the study highlights how intelligent fault diagnosis can support predictive maintenance by enabling earlier detection of bearing defects. Early fault identification can reduce unexpected equipment failures, minimize maintenance costs, improve machine availability and enhance operational safety.

Furthermore, the comparative evaluation of multiple machine learning algorithms provides guidance for selecting appropriate classification techniques for bearing fault diagnosis. The outcomes of this study may also serve as a foundation for future research involving deep learning, real-time condition monitoring systems, sensor fusion or Industrial Internet of Things (IIoT)-based predictive maintenance applications.

CHAPTER TWO

2 LITERATURE REVIEW

2.1 Rolling Element Bearings

2.1.1 Definition of Rolling Element Bearings

Rolling element bearings are precision-engineered machine elements designed to support rotating shafts while minimizing friction between moving components. Unlike plain bearings, which rely on sliding contact, rolling element bearings employ rolling elements such as balls or rollers positioned between an inner race and an outer race. This rolling contact significantly reduces frictional resistance, enhances mechanical efficiency and permits smooth rotational motion under both radial and axial loading conditions

Because of their ability to operate efficiently under varying loads and rotational speeds, rolling element bearings are extensively used in industrial machinery including electric motors, compressors, turbines, pumps, gearboxes, railway systems, wind turbines and manufacturing equipment. Their reliability has a direct influence on machine performance, production efficiency and operational safety. Consequently, bearing health has become one of the most closely monitored aspects of rotating machinery maintenance.

2.1.2 Components of a Rolling Element Bearing

A typical rolling element bearing consists of four primary components:

1. **Inner Race (Inner Ring):** Mounted on the rotating shaft and provides the running surface for the rolling elements.
2. **Outer Race (Outer Ring):** Fixed within the bearing housing and forms the external running surface.
3. **Rolling Elements:** Balls or rollers positioned between the races that reduce friction by converting sliding motion into rolling motion.
4. **Cage (Retainer):** Separates and guides the rolling elements, maintaining uniform spacing and preventing contact between adjacent rolling elements.

The interaction of these components enables efficient load transmission while reducing wear and friction. However, continuous operation under varying mechanical and environmental conditions may eventually lead to fatigue, surface degradation and localized defects that adversely affect bearing performance.

2.1.3 Classification of Rolling Element Bearings

Rolling element bearings can generally be classified according to the geometry of their rolling elements.

2.1.3.1 Ball Bearings

Ball bearings employ spherical rolling elements and are suitable for applications involving moderate radial and axial loads. Their low friction characteristics make them ideal for high-speed rotating machinery.

2.1.3.2 Roller Bearings

Roller bearings utilize cylindrical, tapered, spherical or needle-shaped rollers. Because the contact area between rollers and raceways is larger than that of ball bearings, roller bearings can support substantially higher loads. However, they generally operate at lower rotational speeds than ball bearings.

The choice between ball and roller bearings depends on operating speed, loading conditions, environmental factors and expected service life.

2.1.4 Importance of Rolling Element Bearings in Industry

Rolling element bearings are among the most critical components of rotating machinery because they directly influence machine reliability, energy efficiency and maintenance requirements. Studies have shown that bearing failures account for a significant proportion of failures in electric motors and other rotating equipment, making early fault detection an important objective of industrial maintenance programs.

In modern manufacturing environments, unexpected bearing failures can interrupt production schedules, increase maintenance expenditure and reduce equipment availability. Consequently, industries have increasingly adopted condition monitoring and predictive maintenance techniques to identify bearing defects before catastrophic failure occurs.

2.2 Bearing Failure Mechanisms

2.2.1 Introduction to Bearing Failure

Rolling element bearings are designed to operate reliably under a wide range of loading and environmental conditions. However, like all mechanical components subjected to repeated loading, they are susceptible to gradual deterioration throughout their service life. Bearing failure is typically the result of cumulative damage caused by cyclic stresses, inadequate lubrication, contamination, improper installation, excessive loading or material fatigue. Although bearing failures rarely occur instantaneously, small defects can develop and propagate over time until they significantly impair machine performance or result in complete bearing failure.

The initiation and progression of bearing defects alter the dynamic behaviour of rotating machinery by introducing abnormal vibration patterns, increased noise and temperature variations. As these changes become more pronounced, they provide valuable indicators of bearing health that can be captured using condition monitoring systems. Understanding the mechanisms responsible for bearing failure is therefore fundamental to developing reliable fault diagnosis techniques.

2.2.2 Common Bearing Failure Modes

Bearing failures can arise from several mechanisms, each producing characteristic damage patterns and vibration signatures.

2.2.2.1 Fatigue Failure

Fatigue is the most common cause of rolling element bearing failure. During operation, the repeated contact stresses between the rolling elements and raceways produce microscopic cracks beneath the bearing surface. Over time, these cracks propagate towards the surface, causing material to detach in the form of pits or spalls. Surface fatigue progressively increases vibration levels and reduces bearing performance until replacement becomes necessary.

2.2.2.2 Lubrication Failure

Lubrication is essential for reducing friction, dissipating heat and preventing direct metal-to-metal contact between bearing components. Inadequate lubrication, lubricant contamination or the use of unsuitable lubricants accelerates wear and increases operating temperatures. As lubrication deteriorates, surface damage becomes more severe, reducing bearing lifespan and increasing the likelihood of premature failure.

2.2.2.3 Contamination

The ingress of dust, moisture, metal particles or other foreign materials into a bearing can significantly accelerate wear. Contaminants act as abrasive particles that scratch bearing surfaces and damage the rolling elements and raceways. Even very small particles may initiate microscopic defects that subsequently develop into larger fatigue failures during continued operation.

2.2.2.4 Misalignment

Incorrect shaft alignment causes uneven load distribution across the bearing components. This increases localised stress concentrations and accelerates fatigue damage. Misalignment also produces abnormal vibration patterns, making it one of the conditions that can often be detected through vibration monitoring techniques.

2.2.2.5 Overloading

Bearings are designed to operate within specified load limits. Excessive radial or axial loading increases contact stresses beyond the material's fatigue strength, leading to premature crack initiation, excessive deformation and eventual bearing failure.

2.2.2.6 Improper Installation

Incorrect installation procedures, including excessive mounting force, improper fitting techniques or incorrect bearing seating, may introduce internal stresses or physical damage before the bearing even enters service. Such installation-related defects frequently reduce bearing service life and contribute to premature failure.

2.2.3 Types of Bearing Faults

For vibration-based fault diagnosis, defects are commonly categorised according to their location within the bearing.

2.2.3.1 Inner Race Fault

An inner race fault develops on the rotating inner ring of the bearing. Because the defect passes through the load zone during each shaft revolution, it generates periodic vibration impulses with characteristic frequencies. Inner race defects often produce relatively strong vibration signatures due to the continuous movement of the damaged surface under load.

2.2.3.2 Outer Race Fault

Outer race defects occur on the stationary outer ring. Unlike inner race defects, the damaged location remains fixed relative to the bearing housing. As each rolling element passes over the defect, a series of repetitive impact forces is generated, producing identifiable vibration frequencies that are commonly used in bearing fault diagnosis.

2.2.3.3 Ball (Rolling Element) Fault

Rolling element defects develop on the balls or rollers themselves. These faults produce more complex vibration patterns because the rolling elements simultaneously rotate about their own axes while revolving around the bearing. Consequently, ball defects are often more challenging to identify than raceway defects and may require more advanced feature extraction and classification techniques.

2.2.4 Failure Progression

Bearing deterioration generally follows a progressive sequence rather than occurring suddenly.

Initially, microscopic material defects or surface imperfections develop due to cyclic loading or adverse operating conditions. These defects gradually propagate through repeated mechanical stress, producing increasingly pronounced vibration signatures. As damage progresses, surface pitting develops into larger spalls, increasing vibration amplitude, noise levels and operating temperature. If the bearing continues operating without intervention, severe mechanical degradation eventually leads to loss of rotational accuracy, excessive friction and complete bearing failure.

This gradual progression is the primary reason predictive maintenance strategies are effective. Because measurable changes in vibration occur well before catastrophic failure, condition monitoring systems can identify defects early enough for maintenance personnel to schedule repairs before serious damage occurs.

2.2.5 Relationship Between Bearing Faults and Vibration Analysis

Each bearing defect modifies the vibration behaviour of the rotating system in a unique manner. The location, size and severity of a defect influence the frequency content and statistical characteristics of the resulting vibration signal. Consequently, vibration signals contain valuable diagnostic information that can be extracted through signal processing techniques.

Statistical features such as the mean, standard deviation, root mean square (RMS), skewness, kurtosis and crest factor provide concise numerical descriptions of vibration behaviour and are widely used as inputs to machine learning algorithms. By learning relationships between these extracted features and known fault conditions, machine learning models can accurately classify bearing health states and support intelligent predictive maintenance systems.

This relationship between mechanical fault mechanisms and measurable vibration characteristics forms the theoretical foundation of the methodology adopted in this study.

2.3 Predictive Maintenance and Condition Monitoring

2.3.1.1 Introduction to Maintenance Strategies

Maintenance is a fundamental aspect of industrial operations because it ensures the continuous, safe and efficient operation of machinery. Effective maintenance strategies improve equipment reliability, minimise production downtime and reduce operational costs. As industrial systems have become increasingly automated and complex, maintenance has evolved from reactive repair activities to intelligent condition-based decision-making. The three principal maintenance strategies adopted in industry are corrective maintenance, preventive maintenance and predictive maintenance.

2.3.1.2 Corrective Maintenance

Corrective maintenance, commonly referred to as **run-to-failure maintenance**, involves repairing or replacing equipment only after a failure has occurred. This strategy requires minimal planning and may be economically suitable for inexpensive or non-critical components whose failure has little impact on production.

However, corrective maintenance presents several disadvantages when applied to critical industrial machinery. Unexpected failures often result in unplanned production stoppages, increased repair costs, equipment damage and potential safety hazards. In rotating machinery, undetected bearing failures may propagate to other mechanical components such as shafts, gears and couplings, significantly increasing maintenance costs and repair time.

2.3.2 Preventive Maintenance

Preventive maintenance aims to reduce the likelihood of equipment failure by performing maintenance activities at predetermined intervals based on operating time, production cycles or manufacturer recommendations. Components are inspected, serviced or replaced before the expected end of their service life, regardless of their actual operating condition.

Compared with corrective maintenance, preventive maintenance reduces the probability of catastrophic failures and improves operational reliability. However, because maintenance decisions are based on fixed schedules rather than equipment health, components may be replaced while still capable of operating efficiently. This can lead to unnecessary maintenance expenditure, increased labour costs and reduced equipment availability.

2.3.3 Predictive Maintenance

Predictive maintenance represents a significant advancement over traditional maintenance strategies by basing maintenance decisions on the actual health condition of equipment rather than predetermined schedules. Instead of replacing components at fixed intervals, predictive maintenance continuously or periodically monitors equipment using condition monitoring techniques to identify early signs of deterioration.

When abnormalities are detected, maintenance can be scheduled before catastrophic failure occurs. This approach reduces unnecessary maintenance activities while improving equipment reliability and operational efficiency.

Predictive maintenance offers several important advantages:

- Reduced unplanned downtime.
- Improved machine availability.
- Lower maintenance costs.
- Extended equipment service life.
- Improved workplace safety.
- More efficient utilisation of maintenance resources.

Because predictive maintenance depends on accurate assessment of machine health, reliable condition monitoring techniques are essential for its successful implementation.

2.3.4 Condition Monitoring

Condition monitoring refers to the continuous or periodic assessment of machinery health through the measurement and analysis of physical parameters associated with equipment operation. Rather than waiting for equipment to fail, condition monitoring detects changes in machine behaviour that indicate the development of faults.

Several condition monitoring techniques are widely used in industrial applications, including:

- Vibration analysis
- Acoustic emission analysis
- Thermography
- Oil analysis
- Motor current signature analysis
- Ultrasonic inspection

The selection of an appropriate monitoring technique depends on the machine type, operating environment and failure mechanisms of interest.

Among these techniques, vibration analysis has emerged as one of the most effective methods for monitoring rotating machinery because most mechanical defects directly influence vibration behaviour.

2.3.5 Vibration-Based Condition Monitoring

Rotating machinery naturally produces vibration during normal operation. However, defects such as bearing wear, imbalance, shaft misalignment and gear damage alter both the amplitude and frequency characteristics of these vibrations.

Sensors such as accelerometers are commonly installed on machinery to measure vibration signals. These signals provide valuable information regarding the operating condition of mechanical components and can reveal the presence of faults long before visible damage occurs.

Compared with other condition monitoring techniques, vibration analysis offers several advantages:

- High sensitivity to early-stage bearing defects.
- Non-destructive evaluation.
- Continuous monitoring capability.
- Rapid fault detection.
- Compatibility with automated diagnostic systems.

Because bearing faults produce distinctive vibration signatures, vibration analysis has become the preferred monitoring technique for rolling element bearing fault diagnosis.

2.3.6 Intelligent Predictive Maintenance

Recent developments in sensor technology, Industrial Internet of Things (IIoT), cloud computing and artificial intelligence have transformed predictive maintenance from a manual diagnostic process into an intelligent decision-support system.

Instead of relying solely on expert interpretation, modern predictive maintenance systems utilise machine learning algorithms to analyse large quantities of sensor data automatically. These algorithms identify complex relationships within vibration signals and classify equipment health conditions with high accuracy.

The integration of machine learning into predictive maintenance provides several benefits:

- Automated fault detection.
- Faster diagnostic decisions.
- Improved diagnostic consistency.
- Scalability to large industrial systems.
- Reduced dependence on expert judgement.

These capabilities make machine learning an attractive solution for intelligent condition monitoring, particularly in environments where large volumes of vibration data are generated continuously.

2.3.7 Relevance to the Present Study

The objective of predictive maintenance is not merely to detect faults but to identify them sufficiently early to enable informed maintenance decisions before catastrophic failure occurs. Achieving this objective requires accurate interpretation of condition monitoring data.

In this study, vibration signals obtained from rolling element bearings are processed to extract statistical features that describe the operating condition of the bearing. These features are subsequently used to train supervised machine learning models capable of automatically classifying different bearing fault conditions. By comparing multiple classification algorithms, this research contributes to the development of intelligent vibration-based predictive maintenance systems for rotating machinery.

2.4 Vibration Signal Analysis

2.4.1 Introduction to Vibration Analysis

Vibration analysis is one of the most widely used condition monitoring techniques for assessing the health of rotating machinery. Mechanical components such as bearings, gears, shafts and rotors naturally generate vibration during operation due to the interaction of moving parts. Under normal operating conditions, these vibrations exhibit predictable characteristics that reflect the mechanical condition of the system. However, when defects develop within machine components, the vibration behaviour changes in ways that can be measured and analysed to identify the presence, location and severity of faults.

Because rolling element bearing defects generate distinctive vibration signatures, vibration analysis has become an essential diagnostic tool in predictive maintenance programmes. By monitoring changes in vibration signals over time, maintenance engineers can detect developing faults before catastrophic failure occurs, thereby reducing downtime, maintenance costs and safety risks.

2.4.2 Principles of Machine Vibration

Mechanical vibration refers to the oscillatory motion of a body about its equilibrium position. In rotating machinery, vibration arises from several sources, including shaft rotation, load transmission, structural dynamics and interactions between mechanical components.

A healthy machine produces relatively stable vibration patterns that remain within acceptable operational limits. However, defects such as bearing wear, imbalance, shaft misalignment and looseness introduce additional forces into the mechanical system, altering both the amplitude and frequency content of the vibration signal.

Each mechanical fault generates a characteristic vibration signature because different fault mechanisms produce different excitation forces within the machine. These characteristic signatures provide the basis for vibration-based fault diagnosis.

2.4.3 Sources of Vibration in Rotating Machinery

Several mechanical conditions contribute to vibration generation in rotating equipment, including:

2.4.3.1 Rotor Imbalance

Rotor imbalance occurs when the mass distribution of a rotating component is uneven about its axis of rotation. This condition generates centrifugal forces that increase vibration amplitude at the rotational frequency of the shaft.

2.4.3.2 Shaft Misalignment

Misalignment occurs when the centre lines of connected shafts are not collinear. This produces periodic dynamic forces that increase vibration levels and accelerate wear in bearings and couplings.

2.4.3.3 Mechanical Looseness

Mechanical looseness results from insufficient fastening, worn mounting components or structural degradation. It often produces irregular vibration patterns and impacts during machine operation.

2.4.3.4 Bearing Defects

Defects on the inner race, outer race or rolling elements generate repetitive impact forces as damaged surfaces interact during rotation. These impacts produce characteristic vibration frequencies that can be used to identify specific bearing fault conditions.

Although many machine faults generate vibration, this study focuses specifically on vibration signals associated with rolling element bearing defects.

2.4.4 Vibration Measurement

Machine vibration is typically measured using **accelerometers**, which convert mechanical acceleration into electrical signals. Accelerometers are widely used in industrial condition monitoring because they provide high sensitivity over a broad frequency range and can detect subtle changes associated with early-stage bearing defects.

The measured vibration signal is recorded as a time-varying waveform representing acceleration as a function of time. This raw signal contains information regarding machine behaviour and serves as the primary input for further analysis.

The CWRU Bearing Data Center dataset used in this study consists of vibration signals collected using accelerometers mounted on bearing housings under controlled laboratory conditions. These signals represent both healthy bearings and bearings containing artificially induced defects of varying sizes and locations.

2.4.5 Time-Domain Analysis

Time-domain analysis examines vibration signals directly as they vary with time. It is one of the simplest yet most effective approaches for assessing bearing condition because numerous statistical characteristics can be extracted directly from the waveform.

Common time-domain statistical features include:

- Mean
- Maximum value
- Minimum value
- Standard deviation
- Root Mean Square (RMS)
- Variance
- Skewness
- Kurtosis

- Crest factor
- Waveform factor

These statistical descriptors summarise different aspects of vibration behaviour and provide compact numerical representations suitable for machine learning algorithms.

In this study, statistical features extracted from the time-domain vibration signals constitute the primary input variables for model development.

2.4.6 Frequency-Domain Analysis

While time-domain analysis examines signal amplitude over time, frequency-domain analysis investigates how vibration energy is distributed across different frequencies.

Frequency-domain analysis is commonly performed using the **Fast Fourier Transform (FFT)**, which converts time-domain signals into their corresponding frequency spectra. This transformation enables engineers to identify characteristic defect frequencies associated with specific mechanical faults, including inner race defects, outer race defects and rolling element defects.

Frequency-domain analysis is particularly valuable because many bearing defects produce periodic excitation frequencies that may not be immediately apparent in the raw time-domain waveform.

Although frequency-domain techniques are widely applied in industrial diagnostics, the present study focuses primarily on statistical features extracted from time-domain vibration signals for machine learning-based fault classification.

2.4.7 Importance of Vibration Analysis in Bearing Fault Diagnosis

The effectiveness of vibration analysis lies in its ability to detect defects during the early stages of damage development. As bearing defects evolve, repeated impacts between damaged surfaces generate measurable changes in vibration amplitude and statistical behaviour long before complete mechanical failure occurs.

These changes can be quantified using statistical feature extraction techniques and subsequently analysed using machine learning algorithms capable of distinguishing healthy bearings from various fault conditions.

Consequently, vibration analysis provides the essential link between mechanical fault mechanisms and intelligent diagnostic systems.

2.4.8 Relevance to the Present Study

The methodology adopted in this research is based on the principle that different bearing fault conditions produce distinguishable vibration characteristics. Rather than analysing raw vibration waveforms directly, this study extracts descriptive statistical features from the measured signals to reduce data complexity while preserving important diagnostic information.

These extracted features serve as inputs to supervised machine learning algorithms developed to classify healthy bearings and multiple fault categories. Consequently, vibration signal analysis provides the theoretical basis for the feature extraction, preprocessing and model development procedures presented in Chapter Three.

2.5 Statistical Feature Extraction

2.5.1 Introduction to Feature Extraction

Feature extraction is a fundamental stage in vibration-based fault diagnosis because raw vibration signals are often high-dimensional, noisy and computationally expensive to analyse directly. Rather than using the complete vibration waveform as input to a machine learning model, feature extraction transforms the raw signal into a smaller set of meaningful numerical descriptors that preserve the diagnostic information required for fault classification.

The extracted features provide a compact representation of the vibration signal while reducing data redundancy and computational complexity. Well-selected features enhance the ability of machine learning algorithms to distinguish between healthy bearings and different fault conditions by emphasising characteristics that are most sensitive to changes in machine health.

Feature extraction techniques may be broadly categorised into **time-domain**, **frequency-domain** and **time-frequency-domain** methods. Since this study employs statistical descriptors computed directly from vibration waveforms, the focus is on time-domain statistical feature extraction.

2.5.2 Importance of Statistical Features

Statistical features summarise the characteristics of vibration signals using descriptive numerical measures. These measures provide information about signal magnitude, variability, shape and impulsiveness, all of which change as bearing defects develop.

Compared with raw vibration signals, statistical features offer several advantages:

- Reduced computational complexity.
- Improved model training efficiency.
- Lower memory requirements.
- Better resistance to noise.
- Easier interpretation of machine condition.

- Compatibility with conventional machine learning algorithms.

For these reasons, statistical feature extraction has become one of the most widely adopted approaches for vibration-based bearing fault diagnosis.

2.5.3 Common Statistical Features Used in Bearing Fault Diagnosis

2.5.3.1 Mean

The mean represents the average value of the vibration signal over a specified period.

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

where:

- x_i is the vibration sample,
- N is the total number of samples.

Although the mean alone is not highly sensitive to bearing faults, it provides useful baseline information regarding the overall signal level.

2.5.3.2 Maximum and Minimum Values

The maximum and minimum values indicate the extreme amplitudes present in the vibration signal.

These measures help identify unusually large vibration excursions that may result from severe impacts associated with bearing defects.

2.5.3.3 Standard Deviation

Standard deviation measures the dispersion of vibration amplitudes about the mean.

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N} (x_i - \mu)^2}$$

As bearing defects become more severe, vibration amplitudes typically become more variable, leading to larger standard deviation values.

2.5.3.4 Root Mean Square (RMS)

Root Mean Square (RMS) is one of the most important statistical indicators in vibration analysis because it represents the effective energy contained within the vibration signal.

$$RMS = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i}$$

RMS generally increases as bearing damage progresses and is widely used for machine health assessment in industrial condition monitoring.

2.5.3.5 Variance

Variance measures the average squared deviation of signal amplitudes from their mean.

$$Variance = \frac{1}{N}\sum_{i=1}^{N} (x_i - \mu)^2$$

Variance provides information similar to standard deviation and reflects changes in vibration energy associated with mechanical faults.

2.5.3.6 Skewness

Skewness measures the asymmetry of the vibration amplitude distribution.

Healthy vibration signals often exhibit relatively symmetrical distributions, whereas damaged bearings may generate asymmetric amplitude distributions due to periodic impact events.

2.5.3.7 Kurtosis

Kurtosis measures the sharpness or peakedness of the vibration amplitude distribution.

High kurtosis values frequently indicate impulsive vibration behaviour caused by localised bearing defects such as pitting or spalling.

Because early bearing faults often generate isolated impacts, kurtosis is widely regarded as an effective indicator for incipient defect detection.

2.5.3.8 Crest Factor

Crest Factor is defined as the ratio of the peak signal amplitude to its RMS value.

$$CF = \frac{Peak}{RMS}$$

High crest factor values indicate the presence of impulsive vibration events and are commonly associated with localised bearing defects.

2.5.3.9 Waveform Factor

Waveform Factor compares the RMS value of the vibration signal with its average rectified value.

It provides additional information regarding waveform shape and is useful for distinguishing different fault conditions.

2.5.4 Feature Selection

Although numerous statistical features can be extracted from vibration signals, not all contribute equally to fault diagnosis. Some features provide greater discriminatory power than others, while certain features may contain redundant or irrelevant information.

Feature selection techniques identify the most informative features by evaluating their contribution to classification performance. Reducing the number of input variables simplifies model development, decreases computational requirements and may improve model generalisation by reducing overfitting.

In this study, feature importance analysis was performed to evaluate the contribution of the extracted statistical features prior to machine learning model development.

2.5.5 Relationship Between Statistical Features and Machine Learning

Machine learning algorithms operate on numerical input variables rather than raw vibration waveforms. Consequently, the extracted statistical features constitute the feature space used for classifier training.

Each vibration sample is represented by a vector containing descriptive statistical measures such as RMS, standard deviation, kurtosis, skewness and crest factor. During training, machine learning algorithms learn the relationships between these feature vectors and their corresponding bearing fault classes.

The quality of the extracted features therefore has a direct influence on the classification accuracy of the developed models.

2.5.6 Relevance to the Present Study

This study employs statistical features extracted from vibration signals obtained from the Case Western Reserve University Bearing Data Center dataset. These features form the basis of the machine learning models developed in the subsequent stages of the research.

Following feature extraction, the dataset is preprocessed through feature scaling and label encoding before being used to train and evaluate multiple supervised machine learning algorithms. Feature importance analysis is subsequently conducted to determine the relative contribution of individual statistical descriptors to bearing fault classification.

The extracted feature set provides the essential link between vibration signal analysis and intelligent fault diagnosis, enabling the development of accurate and computationally efficient machine learning models.

2.6 Machine Learning for Bearing Fault Diagnosis

2.6.1 Introduction to Machine Learning

Machine learning is a branch of artificial intelligence that enables computer systems to learn patterns and relationships from data without being explicitly programmed for every task. Rather than relying on predefined rules, machine learning algorithms improve their performance by analysing historical data and making predictions or classifications based on learned patterns.

In recent years, machine learning has become an important tool in mechanical engineering, particularly in predictive maintenance, fault diagnosis, quality control and intelligent manufacturing. The increasing availability of sensor data from industrial equipment has created opportunities for machine learning algorithms to automatically detect abnormalities, classify fault conditions and support maintenance decision-making with minimal human intervention.

For bearing fault diagnosis, machine learning algorithms learn the relationship between extracted vibration features and known bearing conditions. Once trained, these models can classify new vibration data into predefined fault categories, enabling automated condition monitoring and early fault detection.

2.6.2 Types of Machine Learning

Machine learning techniques are generally classified into three main categories.

2.6.2.1 Supervised Learning

Supervised learning uses labelled datasets in which each training sample is associated with a known output or target value. During training, the algorithm learns the relationship between the input features and the corresponding output labels. Once trained, the model predicts the class of previously unseen data.

Because the CWRU dataset contains labelled bearing fault categories, this study adopts supervised learning techniques for fault classification.

2.6.2.2 Unsupervised Learning

Unsupervised learning analyses unlabeled data to identify hidden structures or natural groupings without prior knowledge of the output classes.

Although unsupervised learning is useful for anomaly detection and clustering, it is beyond the scope of this study.

2.6.2.3 Reinforcement Learning

Reinforcement learning enables an intelligent agent to learn optimal actions through interaction with its environment using rewards and penalties.

This learning paradigm is commonly applied in robotics and autonomous systems but is not considered in the present research.

2.6.3 Logistic Regression

Logistic Regression is a supervised classification algorithm that estimates the probability of an observation belonging to a particular class using a logistic (sigmoid) function. Although originally developed for binary classification, it can be extended to multiclass problems using techniques such as One-vs-Rest (OvR) or multinomial logistic regression.

The algorithm is computationally efficient, easy to interpret and often serves as a strong baseline model for classification tasks. However, because it assumes approximately linear decision boundaries, its performance may decline when complex non-linear relationships exist within the data.

In this study, Logistic Regression is employed as the baseline classifier against which more sophisticated machine learning models are compared.

2.6.4 Decision Tree

A Decision Tree is a non-parametric supervised learning algorithm that classifies data by recursively partitioning the feature space into smaller regions using decision rules based on input variables.

The algorithm resembles a flowchart in which each internal node represents a decision based on a feature, each branch represents an outcome of that decision and each terminal node represents a predicted class.

Decision Trees are capable of modelling complex non-linear relationships and provide highly interpretable decision-making processes. However, individual trees are prone to overfitting, particularly when allowed to grow without appropriate constraints.

2.6.5 Random Forest

Random Forest is an ensemble learning algorithm that combines the predictions of multiple Decision Trees to improve classification accuracy and reduce overfitting.

Each tree is trained using a bootstrap sample of the training data while considering only a random subset of features at each split. The final classification is determined through majority voting among all trees in the ensemble.

Random Forest offers several advantages for bearing fault diagnosis:

- High classification accuracy.
- Ability to model non-linear relationships.
- Robustness to noisy data.
- Reduced risk of overfitting.
- Capability to estimate feature importance.

These characteristics make Random Forest particularly suitable for vibration-based fault diagnosis involving multiple statistical features.

2.6.6 K-Nearest Neighbours (KNN)

K-Nearest Neighbours is an instance-based supervised learning algorithm that classifies a new observation according to the classes of its nearest neighbouring samples within the feature space.

The classification depends on the selected number of neighbours (k) and the chosen distance metric, commonly Euclidean distance.

KNN is straightforward to implement and performs well when similar fault conditions produce similar feature patterns. However, its prediction speed decreases for large datasets because all training samples must be considered during classification.

2.6.7 Support Vector Machine (SVM)

Support Vector Machine is a supervised learning algorithm that identifies an optimal decision boundary, known as a hyperplane that maximises the separation between different classes.

For datasets that are not linearly separable, SVM employs kernel functions to project the data into higher-dimensional feature spaces where separation becomes possible.

SVM is particularly effective for high-dimensional datasets and has demonstrated excellent performance in many fault diagnosis applications. Nevertheless, selecting suitable kernel functions and tuning hyperparameters are important factors influencing its performance.

2.6.8 Model Evaluation

The performance of machine learning classifiers is assessed using quantitative evaluation metrics that compare predicted class labels with the true class labels.

The evaluation metrics adopted in this study include:

- **Accuracy:** Measures the proportion of correctly classified observations.

- **Precision:** Indicates the proportion of positive predictions that are correct.
- **Recall:** Measures the ability of the model to correctly identify actual fault classes.
- **F1-Score:** Represents the harmonic mean of precision and recall, providing a balanced measure of classification performance.

These metrics collectively provide a comprehensive assessment of classifier performance, particularly for multiclass classification problems.

2.6.9 Hyperparameter Optimisation

Machine learning models contain parameters learned automatically during training and hyperparameters that must be specified before training begins.

Hyperparameters influence model complexity and predictive performance. Examples include the number of trees in a Random Forest, the maximum tree depth, the number of neighbours in KNN and the regularisation parameters of SVM.

To improve classification performance, this study employs **GridSearchCV** to identify the optimal hyperparameter combination for the best-performing classifier.

2.6.10 Relevance to the Present Study

This study investigates the application of five supervised machine learning algorithms for vibration-based bearing fault diagnosis using statistical features extracted from the CWRU dataset. Each algorithm is trained and evaluated under a common experimental framework to ensure a fair comparison of classification performance.

The evaluation process identifies the classifier that offers the highest predictive accuracy while maintaining strong precision, recall and F1-score performance. The selected model is subsequently optimised through hyperparameter tuning to improve its diagnostic capability.

The knowledge presented in this section forms the theoretical basis for the model development, evaluation and optimisation procedures described in Chapter Three.

2.7 Review of Related Studies

2.7.1 Introduction

The application of machine learning to vibration-based bearing fault diagnosis has received significant attention over the past two decades due to the increasing demand for intelligent predictive maintenance systems. Researchers have investigated various feature extraction techniques, machine learning algorithms and evaluation methods to improve the accuracy and reliability of bearing fault classification. This section reviews selected studies relevant to vibration-based rolling element bearing fault diagnosis, with emphasis on the datasets used, feature extraction methods, classification algorithms and reported performance. The review also identifies limitations in previous research that motivate the present study.

2.7.2 Review of Selected Studies

2.7.2.1 Study 1: Samanta (2004)

Samanta (2004) investigated the use of artificial neural networks and support vector machines for the fault diagnosis of rolling element bearings using vibration signals. Statistical features extracted from vibration measurements were used as input variables for the classifiers. The study demonstrated that machine learning algorithms could effectively distinguish between healthy and faulty bearing conditions and highlighted the importance of feature selection in improving classification performance.

Relevance to the present study:

This study established the feasibility of applying supervised machine learning to vibration-based bearing fault diagnosis. However, it focused on a limited number of algorithms and earlier machine learning techniques. The present study extends this concept by comparing multiple widely used supervised classifiers within a unified experimental framework.

2.7.2.2 Study 2: Widodo and Yang (2007)

Widodo and Yang (2007) presented a comprehensive review of intelligent fault diagnosis methods for rotating machinery, covering support vector machines, artificial neural networks, fuzzy logic and other computational intelligence techniques. The authors concluded that support vector machines demonstrated strong classification performance for vibration-based fault diagnosis, particularly when combined with appropriate feature extraction methods.

Relevance to the present study:

Their review highlights the growing importance of intelligent diagnostic systems and supports the selection of Support Vector Machine as one of the classifiers evaluated in this research.

2.7.2.3 Study 3: Lessmeier et al. (2016)

Lessmeier et al. (2016) introduced the Paderborn University bearing dataset and demonstrated the application of machine learning methods for bearing condition classification under varying operating conditions. The study emphasised the importance of representative datasets and robust evaluation procedures for developing reliable diagnostic models.

Relevance to the present study:

Although this research uses the CWRU dataset rather than the Paderborn dataset, it reinforces the importance of using well-characterised benchmark datasets for evaluating fault diagnosis algorithms.

2.7.2.4 Study 4: Zhang et al. (2019)

Zhang et al. (2019) investigated vibration-based bearing fault diagnosis using statistical feature extraction combined with ensemble learning algorithms. The authors reported that ensemble methods achieved higher classification accuracy than individual classifiers due to their ability to reduce overfitting and improve generalisation performance.

Relevance to the present study:

This finding supports the inclusion of Random Forest as one of the machine learning models evaluated in this research and provides evidence for comparing ensemble and non-ensemble methods.

2.7.2.5 Study 5: Ma et al. (2021)

Ma et al. (2021) explored the application of machine learning techniques for predictive maintenance using vibration-based condition monitoring data. Their work compared several supervised classification algorithms and concluded that model performance depends not only on the selected algorithm but also on data preprocessing, feature engineering and hyperparameter optimisation.

Relevance to the present study:

This observation aligns closely with the methodology adopted in this research, where data preprocessing, feature importance analysis and hyperparameter tuning form integral components of the machine learning workflow.

Table 2- 1 summarises the principal characteristics of the reviewed studies.

Author(s)	Dataset	Features	Algorithm(s)	Key Finding
Samanta (2004)	Experimental vibration data	Statistical features	ANN, SVM	Machine learning effectively classified bearing faults.
Widodo & Yang (2007)	Review of multiple studies	Various	SVM, ANN, Fuzzy Logic	Intelligent methods improved fault diagnosis performance.
Lessmeier et al. (2016)	Paderborn dataset	Statistical and signal-processing features	Multiple ML methods	Benchmark datasets are essential for reliable evaluation.
Zhang et al. (2019)	Bearing vibration data	Statistical features	Ensemble learning	Ensemble methods improved classification accuracy.
Ma et al. (2021)	Condition monitoring data	Statistical features	Multiple supervised algorithms	Preprocessing and hyperparameter tuning strongly influence model performance.

2.7.3 Critical Discussion

The reviewed studies demonstrate that machine learning has become an effective approach for vibration-based bearing fault diagnosis. Most researchers reported high classification performance when informative statistical or signal-processing features were combined with suitable supervised learning algorithms. However, several observations emerge from the literature.

First, classifier performance varies considerably depending on the dataset, extracted features and preprocessing techniques employed. This makes direct comparison between studies difficult.

Second, while many studies focus on achieving high classification accuracy, fewer provide systematic comparisons of multiple algorithms using a consistent preprocessing pipeline and evaluation framework. Such comparisons are important for understanding the relative strengths and weaknesses of different classifiers under identical experimental conditions.

Third, several studies emphasise advanced deep learning approaches. Although these methods can achieve excellent predictive performance, they often require substantially larger datasets, greater computational resources and reduced model interpretability compared with conventional supervised machine learning techniques. In many engineering applications, interpretable models trained on carefully engineered statistical features remain attractive because they balance predictive performance with computational efficiency and practical implementation.

These observations provide the context for the present study, which evaluates multiple supervised machine learning algorithms using the same extracted statistical features, preprocessing procedures and evaluation metrics on the CWRU bearing vibration dataset.

CHAPTER THREE

3 RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes the methodology adopted for the development and evaluation of a machine learning-based bearing fault diagnosis system using vibration signal data. It presents the research design, dataset description, development environment, data collection procedures, exploratory data analysis, data preprocessing techniques, feature importance analysis, machine learning model development, model evaluation methods and hyperparameter optimisation process employed in the study.

The methodology was designed to provide a systematic and reproducible framework for developing supervised machine learning models capable of accurately classifying rolling element bearing fault conditions. Statistical features extracted from vibration signals were utilised as input variables for model development, while standard machine learning evaluation metrics were employed to assess classifier performance.

3.2 Research Design

This study adopted an **experimental research design** to investigate the application of supervised machine learning techniques for rolling element bearing fault diagnosis using vibration signal data. An experimental design was considered appropriate because it enabled the systematic development, training and evaluation of multiple machine learning models under controlled conditions using the same dataset, preprocessing procedures and evaluation criteria.

The research followed a quantitative data-driven approach in which statistical features extracted from vibration signals served as the independent variables, while the bearing fault categories constituted the dependent variable. The study involved the implementation of five supervised machine learning algorithms including: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours and Support Vector Machine, to classify different bearing health conditions.

To ensure fairness in model comparison, all algorithms were trained using identical preprocessing procedures and evaluated using the same train-test split and performance metrics. Following the initial evaluation, the best-performing classifier was further improved through hyperparameter optimisation using GridSearchCV. The final optimised model was subsequently evaluated to determine its suitability for vibration-based bearing fault diagnosis.

3.3 Dataset Description

The dataset used in this study was obtained from the **Case Western Reserve University (CWRU) Bearing Data Center**, one of the most widely recognised benchmark datasets for bearing fault diagnosis research. The dataset contains vibration signals collected from rolling element bearings operating under controlled laboratory conditions with both healthy and artificially induced fault states.

The experimental test rig comprised an electric motor, a torque transducer, a dynamometer and deep groove ball bearings. Bearing defects were introduced using electrical discharge machining (EDM) to create faults on the inner race, outer race and rolling elements. Vibration signals were measured using accelerometers mounted on the drive end and fan end bearing housings while the machinery operated under different load conditions.

For this study, the preprocessed statistical feature dataset derived from the CWRU vibration signals was utilised. The dataset consisted of statistical descriptors extracted from the vibration measurements, including measures of central tendency, variability and signal shape. These features formed the input variables for the supervised machine learning models, while the encoded bearing fault categories served as the target variable.

The use of the CWRU dataset offers several advantages, including its widespread acceptance in bearing fault diagnosis research, high-quality vibration measurements and well-documented fault conditions. These characteristics make it an appropriate benchmark for evaluating the performance of machine learning algorithms.

3.4 Development Environment

The machine learning models developed in this study were implemented using the Python programming language within the Jupyter Notebook environment. Python was selected because of its extensive ecosystem of scientific computing and machine learning libraries, ease of implementation and widespread adoption in data science and engineering research.

Several open-source Python libraries were employed during the implementation of the study. **NumPy** was used for numerical computations and efficient manipulation of multidimensional arrays. **Pandas** was utilised for data loading, cleaning, transformation and tabular data analysis. **Matplotlib** was employed to generate visualisations for exploratory data analysis, feature importance and model evaluation. Machine learning models were developed using the **Scikit-learn** library, which provided implementations of the classification algorithms, preprocessing techniques, evaluation metrics and hyperparameter optimisation methods used in the study.

The development process was carried out using **Visual Studio Code (VS Code)** with the **Jupyter extension**, providing an integrated environment for code development, execution and documentation. Package management and the Python execution environment were configured through the Anaconda distribution, which simplified dependency management and ensured compatibility among the required scientific computing libraries.

The software environment adopted in this study is summarised in the table below:

Table 3- 1 summary of development environment

Software / Library	Purpose
Python	Programming language used for the machine learning development
Visual Studio Code (VS Code)	Integrated development environment (IDE)
Jupyter Notebook	Interactive notebook environment for the experimentation and documentation
Anaconda	Python distribution and package management
Numpy	Numerical computing and array operations
Pandas	Data loading, preprocessing and manipulation
Matplotlib	Data visualization and plotting
Scikit-learn	Machine Learning algorithms, preprocessing, model evaluation and hyperparameter tuning

3.5 Data Collection

3.5.1 Data Source

The dataset used in this study was obtained from the **Case Western Reserve University (CWRU) Bearing Data Center**, a publicly available benchmark dataset extensively used in bearing fault diagnosis and condition monitoring research. The dataset was developed to facilitate the evaluation and comparison of fault diagnosis techniques under controlled laboratory conditions.

The original CWRU experimental setup consisted of a 2-horsepower induction motor connected to a torque transducer and a dynamometer. Deep groove ball bearings were installed at both the drive end and fan end of the motor. Artificial defects of known sizes were introduced into selected bearings using **Electrical Discharge Machining (EDM)** to simulate common bearing faults. During machine operation, vibration signals were collected using accelerometers mounted on the bearing housings under different operating loads and motor speeds.

The availability of labelled vibration signals corresponding to healthy bearings and multiple fault conditions makes the CWRU dataset an appropriate benchmark for evaluating machine learning algorithms for bearing fault diagnosis.

3.5.2 Dataset Used in This Study

Although the original CWRU Bearing Data Center provides raw vibration signals, this study utilised a **feature-engineered dataset** derived from those signals. The dataset was obtained from **Kaggle**, where statistical features had already been extracted from the raw vibration measurements.

The dataset consists of numerical statistical descriptors representing the vibration characteristics of each observation. These features include measures of signal magnitude, variability and distribution, making them suitable inputs for supervised machine learning algorithms.

The target variable represents the bearing health condition, including normal operation and different bearing fault categories. Prior to model development, the target labels were encoded into numerical values to facilitate machine learning classification.

By using a feature-engineered dataset, the study focused on developing, evaluating and optimising machine learning models rather than performing raw vibration signal processing.

3.5.3 Data Acquisition Procedure

The data acquisition procedure adopted in this study consisted of the following steps:

1. The feature-engineered bearing fault dataset was downloaded from Kaggle.
2. The dataset was imported into the Python environment using the Pandas library.
3. The imported dataset was examined to verify its dimensions, feature names and data types.
4. The dataset was inspected for missing values, duplicate records and inconsistencies.
5. Exploratory Data Analysis (EDA) was performed to understand the structure and characteristics of the dataset.
6. Following exploratory analysis, the dataset was preprocessed through label encoding, feature scaling and preparation for machine learning model development.

This structured workflow ensured that the dataset was suitable for supervised learning and provided a consistent foundation for all subsequent experiments.

3.5.4 Dataset Variables

The dataset consists of predictor variables (input features) and one target variable.

The predictor variables are statistical descriptors extracted from vibration signals, including:

- Mean
- Standard Deviation
- Root Mean Square (RMS)
- Variance
- Skewness
- Kurtosis
- Crest Factor
- Waveform Factor
- Maximum Value
- Minimum Value

The target variable represents the bearing condition, consisting of multiple classes corresponding to healthy bearings and different bearing fault types.

During preprocessing, the categorical target labels were encoded into numerical values while preserving the relationship between the encoded values and their original fault categories.

3.5.5 Justification for Dataset Selection

The CWRU-derived dataset was selected for this study for several reasons:

- It is one of the most widely recognised benchmark datasets in bearing fault diagnosis research.
- It contains labelled observations suitable for supervised machine learning.
- The dataset has been extensively validated in previous academic studies, facilitating comparison with existing research.
- The extracted statistical features are computationally efficient and appropriate for conventional machine learning algorithms.
- The dataset provides multiple bearing fault categories, enabling multiclass classification.

These characteristics make the dataset suitable for evaluating the performance of different supervised machine learning algorithms in bearing fault diagnosis.

Table 3- 2 Summary of dataset

Attribute	Description

Dataset Source	Case Western Reserve University (CWRU) Bearing Data Center (accessed via Kaggle)
Dataset Type	Feature-engineered vibration dataset
Problem Type	Multiclass Classification
Input Variables	Statistical vibration features
Target Variable	Bearing Fault Class
Number of Classes	10
Missing Values	None

3.6 Exploratory Data Analysis (EDA)

3.6.1 Introduction

Exploratory Data Analysis (EDA) was conducted to understand the structure, characteristics and quality of the dataset before preprocessing and model development. EDA provides an initial assessment of the data by identifying its dimensions, variable types, class distribution, feature relationships and potential data quality issues such as missing values or duplicate records.

The insights obtained during this stage guided subsequent preprocessing decisions, including label encoding, feature scaling and feature importance analysis.

3.6.2 Dataset Overview

The dataset was first imported into the Python environment using the Pandas library and examined to determine its overall structure. Information such as the number of observations, number of variables, data types and memory usage was obtained using the `info()` and `describe()` functions.

The analysis confirmed that the dataset consisted of statistical features extracted from vibration signals together with a target variable representing the bearing fault class.

Table 3- 3 Dataset summary statistics

Attribute	Value
Number of Samples	2300
Number of Features	9
Target Variable	Fault Class
Feature Types	Numerical
Missing Values	None
Duplicate Records	None

3.6.3 Data Quality Assessment

The dataset was examined for missing values, duplicate records and inconsistent data types before further analysis.

Missing value analysis showed that the dataset contained **no missing observations**, indicating that all statistical features were available for each sample. Duplicate record analysis was also performed to ensure that repeated observations would not bias the machine learning models. Data type verification confirmed that the predictor variables were numerical and suitable for statistical analysis and machine learning.

The absence of significant data quality issues simplified the preprocessing stage and ensured that no data imputation procedures were required.

3.6.4 Descriptive Statistical Analysis

Descriptive statistics were computed to summarise the distribution of the numerical features. Measures including the mean, standard deviation, minimum value, maximum value and quartiles were calculated for each feature.

These statistics provided insight into the scale and variability of the extracted vibration features and revealed differences in feature ranges. The observed variation in feature magnitudes indicated the necessity of feature scaling prior to machine learning model training.

	max	min	mean	sd	rms	skewness	kurtosis	crest	form
count	2300.000000	2300.000000	2300.000000	2300.000000	2300.000000	2300.000000	2300.000000	2300.000000	2300.000000
mean	1.575079	-1.550994	0.015711	0.341601	0.342289	-0.042251	2.664444	4.173130	26.544769
std	1.578422	1.602706	0.006469	0.305279	0.304813	0.180774	4.411096	1.148349	29.209702
min	0.157300	-6.292600	0.003246	0.059140	0.061067	-1.089928	-0.803795	2.428511	3.484429
25%	0.456398	-2.174975	0.011236	0.135506	0.136374	-0.103426	-0.015164	3.260382	7.413359
50%	0.794510	-0.733700	0.013730	0.188551	0.190662	-0.002466	0.816970	3.921650	13.122811
75%	2.278425	-0.426987	0.018638	0.555589	0.555671	0.061093	3.902286	4.815876	39.911894
max	6.825900	-0.160220	0.038386	1.256577	1.256311	1.059512	30.385326	8.821577	313.742612

Figure 3- 1Descriptive Statistics of Numerical Features

3.6.5 Class Distribution

The distribution of the target variable was examined to determine the number of observations belonging to each bearing fault category.

Understanding class distribution is important because severe class imbalance may bias machine learning models toward the majority class. The analysis showed the proportion of samples associated with each fault category, providing an indication of the balance of the dataset.

The class distribution is illustrated in the figure below:

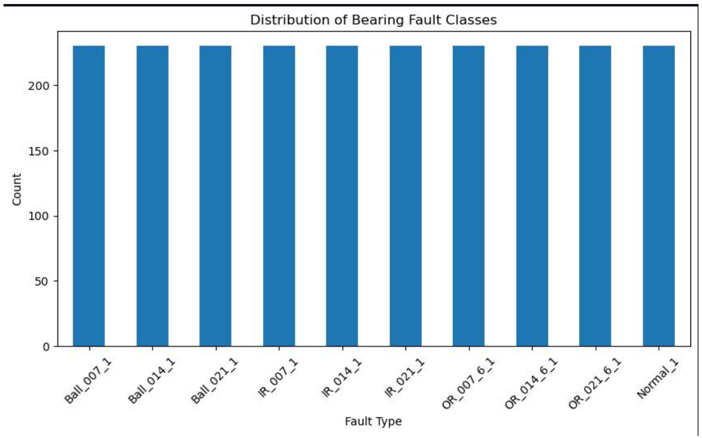

Figure 3- 2 Distribution of bearing fault classes

3.6.6 Feature Correlation Analysis

Correlation analysis was performed to investigate the relationships among the extracted statistical features.

A correlation matrix and corresponding heatmap were generated using Pearson's correlation coefficient. Highly correlated features may contain redundant information and can influence model behaviour, particularly for algorithms that assume feature independence.

The correlation heatmap provided a visual representation of the strength and direction of relationships between pairs of features.

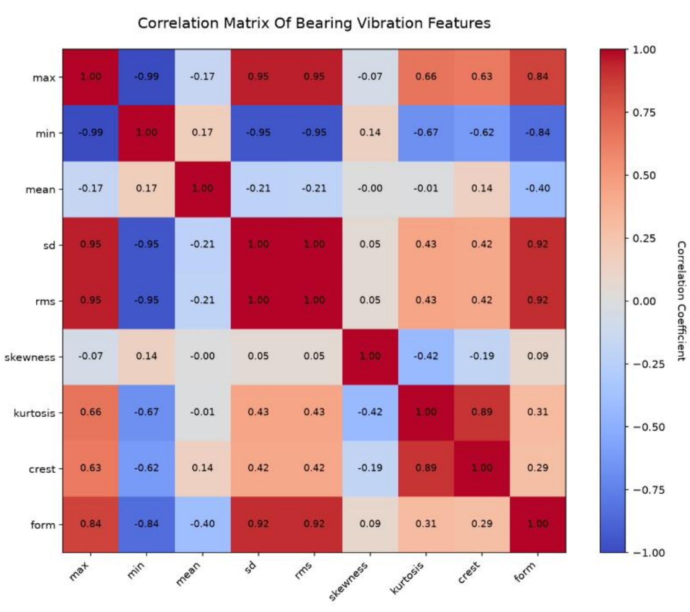

Figure 4- 3Correlation Heatmap of Statistical Features

3.6.7 Feature Distribution

The distributions of the numerical features were visualised using histograms and boxplots to identify skewness, variability and potential outliers.

The analysis revealed that several statistical features exhibited different scales and distribution patterns. Some features also contained extreme values that are characteristic of vibration-based fault diagnosis datasets. Since these observations represent genuine machine behaviour rather than measurement errors, they were retained for further analysis.

The findings reinforced the decision to standardise the predictor variables before model training.

Representative feature distributions are shown in the figure below:

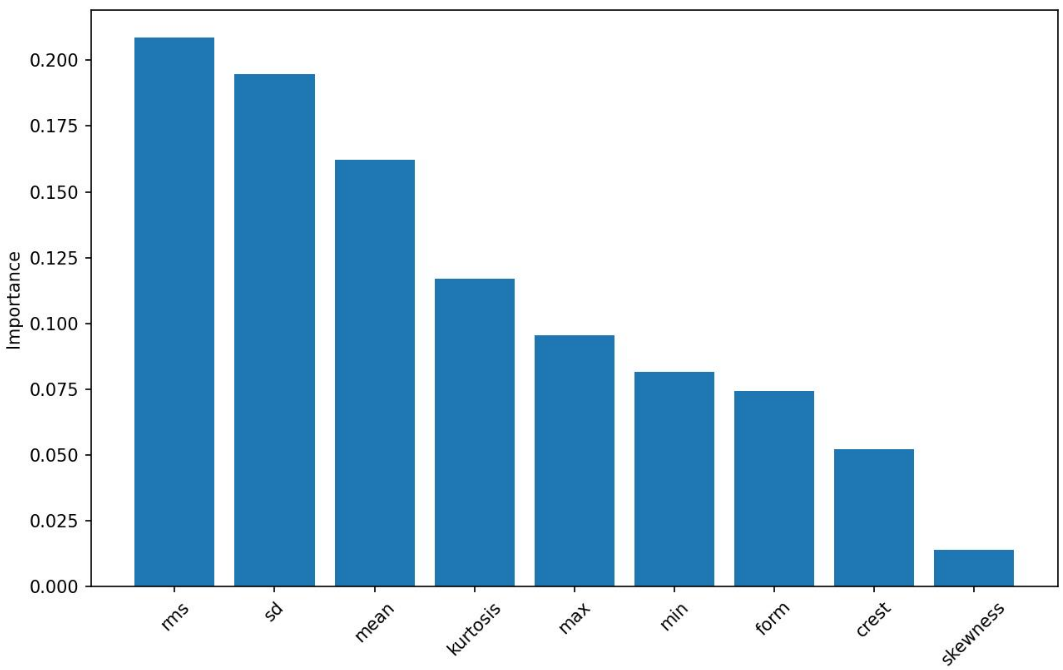

Figure 4- 4Distribution of Selected Statistical Features

3.6.8 Summary of Exploratory Data Analysis

The exploratory data analysis provided valuable insight into the structure and characteristics of the dataset. The analysis confirmed that the dataset was complete, with no missing values or major inconsistencies. Descriptive statistics and visualisations demonstrated that the statistical features varied considerably in scale, supporting the need for feature standardisation during preprocessing.

Furthermore, correlation analysis highlighted relationships among several predictor variables, while examination of the target variable confirmed the suitability of the dataset for supervised multiclass classification. The findings from the exploratory analysis informed the preprocessing procedures described in the following section.

3.7 Data Preprocessing

3.7.1 Introduction

Data preprocessing was performed to prepare the dataset for machine learning model development.

The preprocessing stage involved separating the predictor variables from the target variable, encoding categorical class labels into numerical values, splitting the dataset into training and testing subsets and standardising the predictor variables. These procedures ensured consistency during model training and improved the performance of algorithms sensitive to differences in feature scale.

3.7.2 Feature and Target Separation

The dataset consisted of nine statistical vibration features and one target variable representing the bearing fault class.

The predictor variables comprised:

- Maximum Value (`max`)
- Minimum Value (`min`)
- Mean (`mean`)
- Standard Deviation (`sd`)
- Root Mean Square (`rms`)
- Skewness (`skewness`)
- Kurtosis (`kurtosis`)
- Crest Factor (`crest`)
- Waveform Factor (`form`)

The target variable (`fault`) contained the corresponding bearing condition labels.

The predictor variables were assigned to the feature matrix (**X**), while the target variable was assigned to the response vector (**y**) for subsequent machine learning model development.

3.7.3 Label Encoding

The target variable consisted of categorical fault labels representing different bearing health conditions. Since supervised machine learning algorithms implemented in Scikit-learn require numerical target values, the fault labels were transformed into integer representations using the `LabelEncoder` class.

Each unique bearing condition was assigned a corresponding numerical label while preserving a one-to-one mapping between the original fault categories and their encoded values.

This transformation enabled the classifiers to process the target variable while maintaining the integrity of the class information.

Table 3- 4 Encoded Bearing Fault Classes

Original Fault Label	Encoded Value
Ball_007_1	*0*
Ball_014_1	*1*
Ball_021_1	*2*
IR_007_1	*3*
IR_014_1	*4*
IR_021_1	*5*
Normal_1	*6*
OR_007_6_1	*7*
OR_014_6_1	*8*
OR_021_6_1	*9*

3.7.4 Dataset Partitioning

To evaluate the generalisation capability of the machine learning models, the dataset was divided into training and testing subsets using the `train_test_split()` function from the Scikit-learn library.

A stratified train-test split was employed to preserve the class distribution in both subsets. This approach ensured that each bearing fault category remained proportionally represented in the training and testing datasets, thereby reducing sampling bias during model evaluation.

The training dataset was used to develop the machine learning models, while the testing dataset was reserved exclusively for evaluating predictive performance on previously unseen data.

3.7.5 Feature Standardisation

The extracted statistical features exhibited different numerical ranges and units. Algorithms such as Logistic Regression, K-Nearest Neighbours and Support Vector Machine are particularly sensitive to variations in feature scale.

To ensure that all predictor variables contributed equally during model training, feature standardisation was performed using the `StandardScaler` class from Scikit-learn.

Standardisation transforms each feature by subtracting its mean and dividing by its standard deviation, resulting in variables with approximately zero mean and unit variance.

Importantly, the scaler was **fitted only on the training dataset** and then applied to both the training and testing datasets. This prevented information from the testing dataset from influencing the training process and helped avoid data leakage.

3.7.6 Summary of Data Preprocessing

The preprocessing stage successfully prepared the dataset for supervised machine learning. Categorical fault labels were encoded into numerical values, the dataset was partitioned into training and testing subsets, and feature standardisation was applied to ensure consistent feature scaling.

3.8 Feature Importance Analysis

3.8.1 Introduction

Feature importance analysis was conducted to evaluate the contribution of each statistical vibration feature to bearing fault classification. Although all extracted features contain useful information, their individual predictive capabilities may differ. In addition to improving interpretability, feature importance analysis can guide feature selection and help determine whether certain variables have little influence on model performance

3.8.2 Purpose of Feature Importance Analysis

The objectives of the feature importance analysis were to:

- Evaluate the contribution of each statistical feature to fault classification.
- Identify the most influential vibration characteristics.
- Detect features with relatively low predictive value.
- Improve the interpretability of the machine learning models.
- Provide engineering insight into the statistical characteristics most strongly associated with bearing faults.

3.8.3 Methodology

Feature importance analysis was performed using the preprocessed training dataset generated during the data preprocessing stage. The analysis was conducted before model comparison to gain an understanding of the predictive significance of each statistical feature.

The important values were then ranked from the most influential to the least influential feature and visualised using a horizontal bar chart.

3.8.4 Results of Feature Importance Analysis

The analysis revealed that the statistical vibration features did not contribute equally to the classification task. Certain features exhibited substantially greater predictive importance, indicating that they contained more discriminative information regarding bearing health conditions.

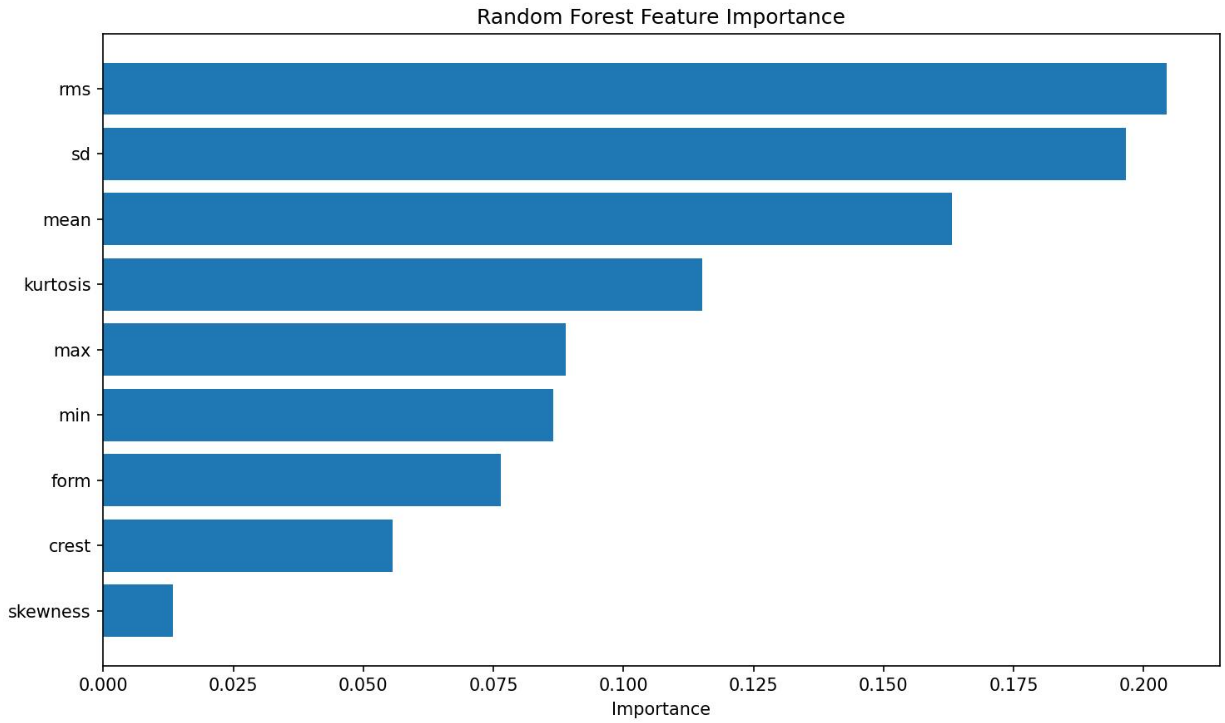

Figure 4- 5 Feature Importance Ranking of Statistical Vibration Features

3.8.5 Discussion

The results indicate that some statistical descriptors contain stronger diagnostic information than others. Features associated with vibration energy, impulsiveness and signal distribution generally provide greater discrimination between healthy and faulty bearings because bearing defects alter the dynamic response of rotating machinery.

The feature importance analysis confirms that the selected statistical features collectively provide sufficient information for supervised machine learning algorithms to distinguish among the different bearing conditions. It also demonstrates that no single feature is solely responsible for classification performance; rather, the combination of complementary vibration characteristics contributes to the predictive capability of the models.

These findings support the use of the complete feature set for subsequent model development and comparison.

3.8.6 Relevance to Model Development

The insights obtained from feature importance analysis informed the next stage of the study, namely the development of supervised machine learning classifiers.

Understanding the relative contribution of each predictor variable provides greater confidence that the selected features are meaningful and supports interpretation of the classification results obtained in Chapter Four.

Furthermore, the analysis establishes a connection between the engineering characteristics of vibration signals and the computational learning process.

3.8.7 Summary

Feature importance analysis demonstrated the relative predictive contribution of the extracted statistical vibration features. The analysis provided a ranked assessment of feature significance and confirmed that the selected feature set was suitable for machine learning-based bearing fault classification.

The results obtained in this stage served as a foundation for the development and comparative evaluation of multiple supervised machine learning algorithms presented in the subsequent section.

3.9 Model Development

3.9.1 Introduction

The objective of the model development stage was to construct supervised machine learning models capable of classifying rolling element bearing conditions using the extracted statistical vibration features. Following data preprocessing and feature importance analysis, multiple classification algorithms were implemented to learn the relationship between the predictor variables and the corresponding bearing fault classes.

To ensure a fair comparison, all models were trained using the same training dataset generated during preprocessing and were evaluated using the same testing dataset. This consistent experimental framework ensured that differences in predictive performance were attributable to the learning algorithms rather than variations in data preparation.

3.9.2 Selection of Machine Learning Models

Five supervised machine learning algorithms were selected for this study based on their popularity in fault diagnosis research, computational efficiency and suitability for multiclass classification problems.

The selected models were:

- Logistic Regression (LR)
- Decision Tree (DT)
- Random Forest (RF)
- K-Nearest Neighbours (KNN)
- Support Vector Machine (SVM)

These algorithms represent a combination of linear, tree-based, ensemble, instance-based and margin-based learning techniques, providing a comprehensive comparison of different classification approaches.

3.9.3 Model Training Procedure

Each machine learning model was trained using the preprocessed training dataset. The predictor variables consisted of the nine statistical vibration features extracted from the dataset, while the encoded bearing fault labels served as the target variable.

The development procedure followed a consistent workflow:

1. Load the preprocessed training dataset.
2. Initialise the machine learning classifier with its default parameters.
3. Train the classifier using the training data.
4. Generate predictions for the testing dataset.
5. Evaluate the model using standard classification metrics.

Applying the same procedure to all models ensured that performance differences reflected the characteristics of the algorithms rather than inconsistencies in implementation.

3.9.4 Logistic Regression Model

Logistic Regression was implemented as the baseline classifier because of its simplicity, efficiency and interpretability. Although it assumes approximately linear decision boundaries, it provides an important reference point for evaluating the performance of more complex algorithms.

The model was trained using the standard implementation provided by the Scikit-learn library and configured for multiclass classification.

3.9.5 Decision Tree Model

The Decision Tree classifier was developed to capture non-linear relationships between the extracted statistical features and the bearing fault classes.

Decision Trees recursively partition the feature space into homogeneous regions using decision rules that maximise class separation. Their interpretability makes them attractive for engineering applications where understanding the decision process is important.

3.9.6 Random Forest Model

Random Forest was implemented as an ensemble learning method consisting of multiple Decision Trees.

Each tree was trained on a bootstrap sample of the training dataset, while random feature selection at each split reduced correlation among individual trees. The final prediction was determined through majority voting.

Because Random Forest combines multiple decision trees, it generally provides improved robustness and reduced overfitting compared with a single Decision Tree.

3.9.7 K-Nearest Neighbours Model

The K-Nearest Neighbours algorithm was included as a distance-based classifier.

Unlike the previous models, KNN does not construct an explicit predictive model during training. Instead, classification is performed by identifying the nearest training samples to each testing observation using a distance metric.

Since KNN relies on distance calculations, the feature standardisation performed during preprocessing was particularly important for this classifier.

3.9.8 Support Vector Machine Model

Support Vector Machine was implemented to evaluate its capability for separating bearing fault classes using optimal decision boundaries.

The algorithm seeks a hyperplane that maximises the separation between different classes. For complex datasets, kernel functions enable non-linear separation by projecting the data into higher-dimensional feature spaces.

Because SVM is sensitive to feature scaling, the standardised dataset generated during preprocessing was used throughout model training.

3.10 Model Evaluation Metrics

3.10.1 Introduction

Following the development of the supervised machine learning models, a systematic evaluation procedure was adopted to assess their predictive performance. Model evaluation is an essential stage in machine learning because it provides objective measures of a classifier's ability to correctly identify bearing fault conditions from previously unseen vibration data.

To ensure a fair comparison among the selected classifiers, all models were evaluated using the same testing dataset and identical performance metrics. The evaluation metrics employed in this study were selected because they provide complementary information regarding different aspects of classification performance.

3.10.2 Confusion Matrix

The Confusion Matrix was used as the primary tool for visualising the classification performance of each machine learning model. It summarises the relationship between the actual bearing fault classes and the corresponding predicted classes generated by the classifier.

For multiclass classification problems such as bearing fault diagnosis, the confusion matrix provides a detailed breakdown of correctly classified observations and misclassifications across all fault categories.

The confusion matrix assists in identifying which fault classes are easily distinguished and which classes are more likely to be confused by the classifier.

3.10.3 Accuracy

Accuracy measures the overall proportion of correctly classified observations relative to the total number of predictions made by the model.

It is expressed as:

$$Accuracy = \frac{Number\ of\ correct\ predictions}{Total\ number\ of\ predictions}$$

Accuracy provides an overall indication of classifier performance and serves as one of the most commonly reported evaluation metrics in supervised machine learning.

3.10.4 Precision

Precision measures the proportion of observations predicted to belong to a particular fault class that are actually members of that class.

It is calculated as:

$$Precision = \frac{TP}{TP + FP}$$

Where:

- TP = True Positives
- FP = False Positives

High precision indicates that the classifier makes relatively few false positive predictions.

In the context of bearing fault diagnosis, high precision implies that when a model predicts a particular fault condition, the prediction is likely to be correct.

3.10.5 Recall

Recall, also known as sensitivity, measures the ability of a classifier to correctly identify all observations belonging to a particular fault class.

It is expressed as:

$$Recall = \frac{TP}{TP + FN}$$

Where:

- TP = True Positives
- FN = False Negatives

A high recall value indicates that the model successfully detects most occurrences of a given bearing fault.

In predictive maintenance applications, recall is particularly important because failing to detect an existing fault may lead to unexpected equipment failure.

3.10.6 F1-Score

The F1-score combines precision and recall into a single performance measure by calculating their harmonic mean.

It is given by:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

The F1-score provides a balanced assessment of classifier performance, particularly when both false positives and false negatives are considered important.

Because bearing fault diagnosis requires accurate identification of all fault categories, the F1-score serves as an important complementary metric to overall accuracy.

3.10.7 Classification Report

The Scikit-learn classification report was used to provide a detailed summary of model performance for each bearing fault class.

The report includes:

- Precision
- Recall
- F1-score
- Support (number of observations in each class)

This detailed evaluation enables the performance of the classifier to be examined for individual fault categories rather than relying solely on overall accuracy.

3.10.8 Comparative Evaluation

To identify the most suitable classifier for bearing fault diagnosis, all five machine learning models were evaluated using the same performance metrics.

The comparative evaluation enabled objective assessment of:

- Overall predictive accuracy
- Ability to distinguish individual bearing fault classes
- Consistency of classification performance
- Suitability for vibration-based fault diagnosis

The classifier demonstrating the strongest overall performance was subsequently selected for hyperparameter optimisation.

3.10.9 Summary

The evaluation framework adopted in this study combined several complementary performance metrics to provide a comprehensive assessment of the developed machine learning models.

The use of accuracy, precision, recall, F1-score, confusion matrices and classification reports ensured that model performance was evaluated from multiple perspectives rather than relying on a single metric. This comprehensive evaluation formed the basis for selecting the most effective classifier for further optimisation.

	Model	Accuracy	Precision	Recall	F1 Score
0	LogisticRegression	0.906522	0.908706	0.906522	0.904648
1	DecisionTreeClassifier	0.932609	0.935377	0.932609	0.932555
2	RandomForestClassifier	0.943478	0.946507	0.943478	0.943773
3	KNeighborsClassifier	0.908696	0.917315	0.908696	0.908978
4	SVC	0.913043	0.922420	0.913043	0.911488

Figure 4- 6 Metric Comparison Table

3.11 Hyperparameter Optimisation

3.11.1 Introduction

Following the comparative evaluation of the baseline machine learning models, the best-performing classifier was selected for further improvement through hyperparameter optimisation. While machine learning algorithms can be trained using default parameter settings, these default values may not provide optimal predictive performance for a specific dataset.

Hyperparameter optimisation was therefore performed to identify the combination of parameter values that produced the highest classification performance while maintaining good generalisation to unseen data.

3.11.2 Purpose of Hyperparameter Optimisation

The primary objectives of hyperparameter optimisation were to:

- Improve the predictive performance of the selected classifier.
- Reduce the possibility of overfitting or underfitting.
- Identify the optimal parameter configuration for the dataset.
- Enhance the model's generalisation capability.
- Produce a more reliable classifier for bearing fault diagnosis.

Rather than relying on trial-and-error parameter selection, a systematic search strategy was adopted to ensure an objective and reproducible optimisation process.

3.11.3 Hyperparameter Search Method

Hyperparameter optimisation was performed using the **GridSearchCV** function provided by the Scikit-learn library. GridSearchCV performs an exhaustive search over a predefined set of hyperparameter values by training and evaluating multiple model configurations.

For each candidate parameter combination, the model was trained and evaluated using cross-validation. The parameter set that achieved the best average cross-validation performance was selected as the optimal configuration.

This systematic approach ensured that the final model was developed using parameters supported by empirical evaluation rather than arbitrary selection.

3.11.4 Cross-Validation

To improve the reliability of model selection, **k-fold cross-validation** was employed during the optimisation process.

In k-fold cross-validation, the training dataset is divided into **5** equally sized subsets (folds). During each iteration:

1. One fold is reserved for validation.
2. The remaining folds are used for training.

3. The process is repeated until every fold has served as the validation set.
4. The average performance across all folds is computed.

This procedure provides a more robust estimate of model performance and reduces the likelihood of selecting a parameter combination that performs well only on a particular train-test split.

3.11.5 Hyperparameter Grid

A predefined parameter grid was constructed for the selected classifier. The optimisation algorithm evaluated all possible parameter combinations within this grid to determine the configuration that produced the highest cross-validation performance.

The exact parameters included in the search depended on the selected machine learning algorithm.

For example, the **Random Forest** classifier was selected, the parameter grid includes:

- Number of trees (`n_estimators`)
- Maximum tree depth (`max_depth`)
- Minimum samples required to split a node (`min_samples_split`)
- Minimum samples required at a leaf node (`min_samples_leaf`)
- Number of features considered at each split (`max_features`)

3.11.6 Selection of the Best Model

Upon completion of the Grid Search process, the parameter combination that achieved the highest average cross-validation score was selected as the optimal model configuration.

The optimised classifier was subsequently retrained using the complete training dataset before being evaluated on the testing dataset.

This ensured that the final model benefited from all available training data while preserving the independence of the testing dataset for performance assessment.

3.11.7 Saving the Optimised Model

The optimised machine learning model was saved for subsequent prediction and evaluation.

Saving the trained model offers several advantages:

- Eliminates the need for repeated model training.
- Ensures consistent predictions.
- Facilitates deployment in future predictive maintenance applications.
- Improves the reproducibility of the research.

The model was stored using Python's model serialisation tools, allowing it to be reloaded for future use without repeating the optimisation process.

3.11.8 Summary

Hyperparameter optimisation improved the selected machine learning classifier by identifying the most suitable parameter configuration through an exhaustive Grid Search and cross-validation procedure.

The optimised model developed during this stage served as the final classifier evaluated in the subsequent phase of the study. This process enhanced the reliability and predictive capability of the machine learning system while ensuring a systematic and reproducible optimisation methodology.

3.12 Final Model Development

3.12.1 Introduction

Following hyperparameter optimisation, the machine learning model that demonstrated the best overall performance was selected as the final classification model for this study. The optimised model was retrained using the optimal hyperparameter configuration identified during the GridSearchCV process and subsequently evaluated using the testing dataset.

The purpose of this stage was to produce a robust and reliable classifier capable of accurately identifying rolling element bearing fault conditions from statistical vibration features. The final model represents the culmination of the preprocessing, feature analysis, model development and optimisation stages described in the preceding sections.

3.12.2 Development of the Final Model

The best-performing machine learning algorithm, as determined through comparative evaluation and hyperparameter optimisation, was selected for final deployment.

The model was retrained using the complete training dataset and the optimal hyperparameter values obtained from the optimisation process. Retraining the model using all available training data ensured that it fully utilised the information contained within the dataset while preserving the testing dataset for independent performance evaluation.

The resulting classifier served as the final predictive model employed throughout the remainder of the study.

3.12.3 Prediction Using the Final Model

After training, the optimised model was used to generate predictions for the testing dataset.

Each observation in the testing dataset consisted of nine statistical vibration features extracted from the original bearing vibration signals. Based on these features, the trained classifier predicted the corresponding bearing fault category.

The predicted labels were subsequently compared with the actual fault classes to evaluate the classification performance of the final model.

3.12.4 Model Persistence

To improve reproducibility and facilitate future use, the final trained model was saved after successful training.

Persisting the model provides several practical advantages:

- It eliminates the need to retrain the classifier before making future predictions.
- It enables rapid deployment in predictive maintenance systems.
- It preserves the exact model used in this study.
- It supports reproducible research by ensuring that identical predictions can be generated from the saved model.

The model was serialised using Python's model persistence utilities, allowing it to be reloaded for future prediction without repeating the entire training process.

3.13 3.12.5 Prediction Workflow

The operational workflow of the final machine learning model is illustrated below:

```
New Bearing Data
        │
        ▼
Extract Statistical Features
        │
        ▼
Apply Saved StandardScaler
        │
        ▼
Load Trained Model
        │
        ▼
Generate Prediction
        │
        ▼
Predicted Bearing Condition
```

3.13.1 Practical Application

The final machine learning model developed in this study has potential applications in industrial predictive maintenance systems.

By analysing statistical features extracted from vibration measurements, the classifier can assist maintenance engineers in automatically identifying bearing health conditions before catastrophic failure occurs. Such intelligent diagnostic systems may contribute to:

- Reduced unplanned machine downtime.
- Lower maintenance costs.
- Improved equipment reliability.
- Enhanced operational safety.
- More effective condition-based maintenance strategies.

Although this study was conducted using the CWRU benchmark dataset, the methodology can be adapted to industrial vibration monitoring systems with appropriate feature extraction and validation.

3.13.2 Reproducibility of the Workflow

To ensure transparency and reproducibility, the complete machine learning pipeline was implemented using a structured workflow.

The workflow included:

1. Exploratory Data Analysis.
2. Data Preprocessing.
3. Feature Importance Analysis.
4. Baseline Model Development.
5. Model Comparison.
6. Hyperparameter Optimisation.
7. Final Model Development.

Each stage produced outputs that served as inputs for the subsequent stage, creating a modular and reproducible machine learning pipeline.

3.13.3 Summary

The final machine learning model was successfully developed by retraining the best-performing classifier using its optimal hyperparameter configuration. The model was saved for future use and prepared for predictive evaluation using previously unseen testing data.

The structured development process ensured that the final classifier was both reproducible and suitable for vibration-based bearing fault diagnosis. The next chapter presents the experimental results obtained from evaluating the developed machine learning models.

3.14 Legal, Ethical and Professional Implications of AI in Bearing Fault Diagnosis Using Machine Learning

Although the work primarily focuses on improving predictive maintenance, deploying such an AI system in real industrial environments raises several legal, ethical, and professional considerations. These issues concern the responsible use of AI, protection of industrial data, accountability for AI-assisted decisions, and ensuring that the system remains reliable, transparent, and secure.

3.14.1 Data Privacy

The project uses the publicly available **Case Western Reserve University (CWRU)** bearing dataset, which consists of vibration signals collected from machinery under controlled laboratory conditions. The dataset does not contain personal information such as employee identities or customer records.

However, in real-world industrial deployment, the AI system could be integrated with maintenance management systems that store:

- names of maintenance engineers,
- machine operators,
- maintenance schedules,
- production records,
- equipment ownership information.

In such situations, organizations must ensure that personal and operational data are collected only when necessary, stored securely, protected from unauthorized access, and processed in accordance with applicable data protection laws. Maintaining data privacy protects employees' rights, preserves organizational confidentiality, and builds trust in AI-assisted maintenance systems.

3.14.2 General Data Protection Regulation (GDPR)

The project does not explicitly reference the GDPR because the research uses a public vibration dataset rather than personal information. Nevertheless, if the developed AI system were deployed in industries operating within the European Union or processing data relating to EU residents, GDPR obligations would apply.

Organizations would be required to:

- process personal data lawfully and transparently;
- collect only the data necessary for maintenance purposes (data minimization);
- inform individuals about how their data will be used;
- implement appropriate technical and organizational security measures;
- allow individuals to exercise their rights, including access, correction, and deletion of personal data.

Thus, while GDPR is not directly applicable to the experimental dataset, it becomes highly relevant during commercial deployment where employee-related information is processed.

3.14.3 HIPAA (Healthcare)

HIPAA (Health Insurance Portability and Accountability Act) is a United States law that protects patients' health information.

The bearing fault diagnosis project does **not** involve healthcare data, medical devices, or patient information. Consequently, HIPAA is **not directly applicable** to this research.

However, if similar machine learning techniques were adapted for predicting failures in hospital equipment such as MRI scanners or ventilators, any associated patient information would need to comply with HIPAA requirements regarding confidentiality, integrity, and security of health records.

3.14.4 UK Data Protection Act

The UK Data Protection Act 2018 complements the UK's data protection framework and governs the processing of personal data.

Like GDPR, this legislation is not directly applicable to the vibration dataset used in the study because no personal information was processed. However, if the predictive maintenance system were deployed within UK industries and integrated with employee databases or operational records containing personal information, compliance with the UK Data Protection Act would become necessary.

Organizations would need to ensure lawful processing, data security, transparency, and accountability throughout the AI system's lifecycle.

3.14.5 Fairness

Fairness requires that an AI system make decisions consistently and objectively.

In this project, fairness relates to the model's ability to classify all bearing conditions accurately without consistently favoring one fault class over another. The AI should identify healthy bearings, inner race faults, outer race faults, and ball faults with comparable reliability rather than producing systematically poorer results for one class.

Maintaining fairness improves confidence in maintenance decisions and reduces the likelihood of unnecessary repairs or overlooked equipment failures.

3.14.6 Bias

Bias occurs when an AI model produces systematically inaccurate predictions for certain classes because of limitations in the training data or learning process.

Possible sources of bias within this project include:

- imbalanced numbers of fault samples,
- incomplete representation of operating conditions,
- laboratory data that may not fully represent real industrial environments.

The project helps reduce bias by comparing five different machine learning algorithms and selecting the model that demonstrated the highest overall predictive performance.

Nevertheless, future deployment should validate the model using real industrial datasets to ensure consistent performance across different machines and operating environments.

3.14.7 Discrimination

Traditional discrimination based on human characteristics such as race, gender, or religion is not relevant because the AI system classifies machine conditions rather than people.

However, technical discrimination could occur if the model consistently misclassifies one category of bearing fault while accurately detecting others. Such unequal performance may result in:

- unnecessary replacement of healthy bearings,
- failure to detect serious faults,
- increased maintenance costs,
- reduced operational safety.

Regular model evaluation is therefore necessary to ensure balanced performance across all fault categories.

3.14.8 Protected Attributes

Protected attributes generally include characteristics such as age, gender, race, ethnicity, religion, disability, and nationality.

These attributes are absent from the project's dataset because the model analyses vibration signals rather than personal information. Consequently, issues relating to discrimination against protected groups do not arise within the current research.

If future industrial implementations combine machine monitoring with employee information, organizations must ensure that protected attributes do not influence maintenance or employment decisions unfairly.

3.14.9 Transparency

Transparency means that users should understand how the AI system operates and how decisions are produced.

The project promotes transparency by documenting:

- the source of the vibration dataset,
- preprocessing procedures,
- statistical feature extraction,
- machine learning algorithms used,
- model evaluation metrics,
- optimization techniques,
- final performance results.

Such documentation enables researchers and engineers to reproduce the methodology and evaluate the reliability of the developed system.

3.14.10Explainable AI (XAI)

The project does not explicitly implement Explainable AI methods.

Nevertheless, explainability is important in industrial maintenance because engineers must understand why the model predicts that a bearing is healthy or faulty.

Future improvements could incorporate feature importance analysis or other Explainable AI techniques to identify which statistical vibration features contributed most to each prediction. Such explanations would increase user confidence and facilitate troubleshooting.

3.14.11Interpretability

Interpretability refers to how easily humans can understand the internal decision-making process of a machine learning model.

Among the algorithms evaluated:

- Logistic Regression and Decision Trees are relatively easy to interpret.
- Random Forest provides superior predictive performance but is less interpretable because it combines numerous decision trees into a single ensemble model.

The project prioritizes predictive accuracy while acknowledging that more complex models may sacrifice interpretability.

3.14.12Accountability

Despite the high predictive accuracy achieved by the optimized Random Forest model, responsibility for maintenance decisions remains with human professionals.

If incorrect AI predictions lead to equipment damage or production losses, accountability rests with:

- system developers,
- maintenance engineers,
- plant operators,
- organizations deploying the system.

The AI serves as a decision-support tool rather than an autonomous decision-maker.

3.14.13Human Oversight

The project demonstrates that the optimized Random Forest model achieved approximately **94.57% accuracy**, but this also means that prediction errors remain possible.

Human oversight is therefore essential. Engineers should review AI-generated fault predictions before replacing expensive machine components or shutting down industrial equipment. Combining AI recommendations with professional engineering judgment reduces operational risks.

3.14.14Model Governance

Model governance involves managing the AI system throughout its lifecycle.

For this project, effective governance would include:

- documenting model development;
- maintaining version control;
- monitoring predictive accuracy after deployment;
- retraining the model using updated vibration data;
- validating performance periodically;
- recording maintenance decisions influenced by the AI system.

These practices help ensure that the model remains accurate and reliable over time.

3.14.15 Security

The trained model and associated vibration data should be protected against unauthorized access, tampering, and theft.

Unauthorized modification of the model could lead to incorrect fault diagnoses, unnecessary maintenance activities, or catastrophic machine failures.

Security measures should include:

- authentication,
- encryption,
- secure servers,
- controlled user access,
- regular software updates.

3.14.16 Data Protection

Although the research dataset is publicly available, industrial implementations would likely involve confidential operational information.

Organizations should therefore:

- encrypt stored data;
- implement access controls;
- maintain secure backups;
- monitor data usage;
- comply with relevant data protection legislation.

Strong data protection safeguards both organizational assets and employee information.

3.14.17 Cybersecurity

Industrial AI systems are increasingly connected through industrial networks and Internet of Things (IoT) devices.

Cybersecurity measures are therefore essential to protect against:

- hacking,
- malware,
- ransomware,
- unauthorized remote access,
- manipulation of sensor readings.

Compromised sensor data could result in inaccurate fault predictions and potentially unsafe maintenance decisions.

3.14.18 Model Robustness

Model robustness refers to the AI system's ability to maintain reliable performance under varying operating conditions.

A robust bearing diagnosis model should continue producing accurate predictions despite:

- noisy vibration signals,
- varying machine loads,
- changing rotational speeds,
- minor sensor faults.

The project's comparison of multiple algorithms and optimization of the Random Forest model contributes to improving robustness, although additional validation using real industrial datasets would further strengthen confidence in practical deployment.

3.14.19Sustainability

The AI-based predictive maintenance approach contributes positively to industrial sustainability by:

- reducing unnecessary replacement of healthy bearings;
- extending equipment lifespan;
- minimizing material waste;
- reducing unexpected machine failures;
- improving maintenance efficiency;
- lowering production downtime.

These benefits support more sustainable industrial operations and better resource utilization.

3.14.20Computational Cost

The project evaluates algorithms with different computational requirements.

Simpler algorithms such as Logistic Regression require relatively low computational resources, whereas Random Forest requires greater processing power because it constructs multiple decision trees. Hyperparameter optimization further increases computational requirements during model training.

Organizations should therefore balance prediction accuracy with available computing resources when selecting an AI model for deployment.

3.14.21Energy Consumption

Training machine learning models requires computational resources that consume electrical energy.

Although the models used in this project are considerably less energy-intensive than modern deep learning architectures, repeated model training, optimization, and large-scale industrial deployment still contribute to energy consumption. Efficient model design can therefore reduce operational costs while supporting sustainable computing practices.

3.14.22 Environmental Impact

The project has both positive and negative environmental implications.

Positive impacts include:

- reduced equipment waste through early fault detection;
- longer bearing service life;
- fewer emergency replacements;
- improved maintenance efficiency;
- reduced consumption of replacement materials.

Potential negative impacts arise from the electricity consumed during model training and operation. However, because the project relies on relatively lightweight machine learning algorithms rather than computationally intensive deep learning models, its environmental footprint is comparatively modest.

CHAPTER FOUR

4 RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents and discusses the results obtained from the development and evaluation of the machine learning models for rolling element bearing fault diagnosis. The objective of the experimental analysis was to assess the ability of the selected supervised machine learning algorithms to accurately classify bearing health conditions using statistical vibration features extracted from the Case Western Reserve University (CWRU) bearing dataset.

The chapter begins by presenting the performance of the baseline machine learning models developed using their default parameter settings. A comparative evaluation of the classifiers is then carried out using standard performance metrics, including accuracy, precision, recall, F1-score, confusion matrices and classification reports. The results of the feature importance analysis are also discussed to provide insight into the contribution of individual vibration features to the classification process.

Subsequently, the chapter presents the results of the hyperparameter optimisation process, highlighting the improvements achieved through systematic parameter tuning using GridSearchCV. Finally, the performance of the optimised model is analysed and its suitability for bearing fault diagnosis is discussed in relation to predictive maintenance applications.

The findings presented in this chapter provide the basis for evaluating the effectiveness of the proposed machine learning framework and identifying the most suitable classifier for vibration-based bearing fault diagnosis.

4.2 Baseline Model Performance

4.2.1 Introduction

The first stage of the experimental evaluation involved assessing the performance of the baseline machine learning models. Each classifier was trained using the preprocessed training dataset with its default hyperparameter settings and evaluated using the same testing dataset.

The objective of this evaluation was to establish a common performance benchmark before hyperparameter optimisation. Using identical training and testing datasets ensured that any differences in performance were attributable to the learning algorithms rather than variations in data preparation.

The five classifiers evaluated were:

- Logistic Regression (LR)
- Decision Tree (DT)

- Random Forest (RF)
- K-Nearest Neighbours (KNN)
- Support Vector Machine (SVM)

The performance of each model was assessed using accuracy, precision, recall, F1-score, confusion matrices and classification reports.

4.2.2 Logistic Regression Performance

The Logistic Regression classifier served as the baseline linear model for this study. After training on the preprocessed dataset, the model was evaluated using the testing dataset.

The evaluation focused on its ability to correctly classify the ten bearing fault conditions and establish a reference point for comparison with the more advanced non-linear and ensemble learning algorithms.

Table 4- 1 Logistic Regression performance

Metric	Value
Accuracy	90.65%
Precision	90.87%
Recall	90.65%
F1-score	90.46%

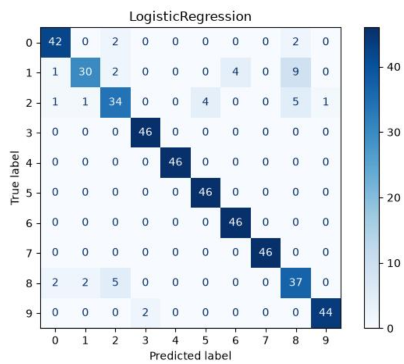

Figure 5- 1 Logistic Regression Confusion Metrics

4.2.2.1 Discussion

The Logistic Regression classifier achieved an **accuracy of 90.65%**, indicating that it correctly classified approximately nine out of every ten bearing fault instances in the testing dataset. The precision score of **90.87%** shows that the model produced relatively few false-positive classifications, while the recall value of **90.65%** demonstrates its ability to correctly identify the majority of the bearing fault conditions present in the dataset.

Similarly, the **F1-score of 90.46%** reflects a balanced trade-off between precision and recall, suggesting that the classifier maintained consistent predictive performance across the ten bearing fault classes.

Although these results indicate good classification capability, the overall performance also highlights the limitations of a linear decision boundary when applied to vibration-based fault diagnosis. The statistical vibration features extracted from rolling element bearings are likely to exhibit complex and non-linear relationships that cannot always be captured effectively by a linear model. Consequently, while Logistic

Regression provides a strong baseline, more advanced algorithms are expected to achieve higher predictive performance.

Overall, the Logistic Regression model establishes a reliable benchmark for subsequent comparison with Decision Tree, Random Forest, K-Nearest Neighbours and Support Vector Machine classifiers.

4.3 Comparative Analysis of Machine Learning Models

4.3.1 Introduction

Following the individual evaluation of the baseline machine learning classifiers, a comparative analysis was conducted to determine the relative performance of each algorithm in classifying rolling element bearing fault conditions. All models were trained using the same preprocessed training dataset and evaluated on the same testing dataset, ensuring a fair and objective comparison.

The classifiers were compared using five evaluation metrics: accuracy, precision, recall, F1-score and confusion matrices. This comprehensive evaluation made it possible to identify the classifier that offered the best overall predictive performance for vibration-based bearing fault diagnosis.

4.3.2 Comparative Performance of the Machine Learning Models

The overall performance of the five baseline machine learning models is presented below:

Table 4- 2 Comparative Performance of the Machine Learning Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Logistic Regression	90.65	90.87	90.65	90.46
Decision Tree	93.26	93.54	93.26	93.26
Random Forest	94.35	94.65	94.35	94.38
K-Nearest Neighbours	90.87	91.73	90.87	90.90
Support Vector Machine	91.30	92.24	91.30	91.15

4.3.3 Accuracy Comparison

As shown in **Table 4.3.2**, the **Random Forest classifier** achieved the highest classification accuracy of **94.35%**, outperforming all other baseline models. The **Decision Tree classifier** ranked second with an accuracy of

93.26%, while the **Support Vector Machine**, **K-Nearest Neighbours** and **Logistic Regression** achieved accuracies of **91.30%**, **90.87%** and **90.65%**, respectively.

The superior performance of the Random Forest classifier can be attributed to its ensemble learning strategy, which combines the predictions of multiple decision trees to reduce variance and improve generalisation. In contrast, Logistic Regression, being a linear classifier, was less effective in modelling the complex relationships between the statistical vibration features and the bearing fault categories.

4.3.4 Precision Comparison

Among the evaluated models, the **Random Forest classifier** achieved the highest precision (**94.65%**), indicating that it generated the fewest false-positive predictions. The **Decision Tree** followed closely with a precision of **93.54%**, while the **Support Vector Machine**, **K-Nearest Neighbours** and **Logistic Regression** achieved precision scores of **92.24%**, **91.73%** and **90.87%**, respectively.

The consistently high precision values across all classifiers demonstrate that the extracted statistical vibration features effectively distinguish between the different bearing fault conditions.

4.3.5 Recall Comparison

The **Random Forest classifier** again demonstrated the highest recall (**94.35%**), indicating that it successfully detected the largest proportion of bearing fault instances. The **Decision Tree classifier** achieved a recall of **93.26%**, followed by the **Support Vector Machine (91.30%)**, **K-Nearest Neighbours (90.87%)** and **Logistic Regression (90.65%)**.

4.3.6 F1-Score Comparison

The results show that the **Random Forest classifier** achieved the highest F1-score (**94.38%**), indicating the best balance between false-positive and false-negative predictions. The **Decision Tree classifier** followed with **93.26%**, while the **Support Vector Machine**, **K-Nearest Neighbours** and **Logistic Regression** recorded F1-scores of **91.15%**, **90.90%** and **90.46%**, respectively.

These findings reinforce the conclusion that the Random Forest model provided the most consistent overall classification performance among the evaluated baseline algorithms.

4.3.7 Overall Ranking of the Machine Learning Models

Based on the combined evaluation metrics, the machine learning models were ranked as follows:

Table 4- 3 Overall Ranking of the Machine Learning Models

Rank	Model	Overall Performance

1	Random Forest	Best overall performance
2	Decision Tree	Very high performance
3	Support Vector Machine	Good performance
4	K-Nearest Neighbours	Good Performance
5	Logistic Regression	Strong baseline Model

4.3.8 Engineering Discussion of the Results

The experimental results demonstrate that all five supervised machine learning algorithms achieved classification accuracies exceeding **90%**, confirming that the selected statistical vibration features are highly effective for rolling element bearing fault diagnosis.

However, the superior performance of the Random Forest classifier indicates that ensemble learning provides a significant advantage when modelling complex vibration characteristics. By aggregating the predictions of multiple decision trees, Random Forest reduces model variance and improves generalisation, leading to more accurate and reliable fault classification.

The Decision Tree classifier also produced strong results, highlighting the suitability of tree-based approaches for fault diagnosis. Nevertheless, as a single-tree model, it is generally more susceptible to overfitting than Random Forest, which explains its slightly lower performance.

The comparatively lower performance of Logistic Regression suggests that a linear decision boundary is insufficient to fully capture the non-linear relationships present in vibration-derived statistical features. Similarly, while K-Nearest Neighbours and Support Vector Machine produced competitive results, they did not surpass the performance of the ensemble learning approach.

Overall, the findings demonstrate that **Random Forest is the most suitable baseline classifier** for the CWRU-derived bearing fault dataset used in this study and therefore represents the strongest candidate for further optimisation through hyperparameter tuning.

4.4 Hyperparameter Optimisation Results

4.4.1 Introduction

Following the comparative evaluation of the baseline machine learning models, the Random Forest classifier was selected for hyperparameter optimisation because it achieved the highest overall classification performance among the five evaluated algorithms. The objective of the optimisation process was to determine whether systematic tuning of the model's hyperparameters could further improve its predictive capability.

Hyperparameter optimisation was performed using the GridSearchCV algorithm provided by the Scikit-learn library. The optimisation process evaluated multiple combinations of parameter values using five-fold cross-validation, ensuring that the selected model configuration generalised well to unseen data.

4.4.2 Hyperparameter Optimisation Strategy

An exhaustive grid search was conducted to identify the optimal configuration of the Random Forest classifier. The optimisation procedure systematically evaluated combinations of the model's key hyperparameters using **5-fold cross-validation**, with **classification accuracy** adopted as the optimisation criterion.

The search considered parameters affecting model complexity and generalisation, including:

- Number of decision trees (`n_estimators`)
- Maximum tree depth (`max_depth`)
- Minimum samples required to split an internal node (`min_samples_split`)
- Minimum samples required at a leaf node (`min_samples_leaf`)

The parameter combination producing the highest mean cross-validation accuracy was selected as the optimal configuration.

4.4.3 Optimal Hyperparameters

The GridSearchCV process identified the following optimal hyperparameter values for the Random Forest classifier.

Table 4- 4 Optimal Hyperparameters for the Random Forest Classifier

Hyperparameter	Optimal Value
Number of Trees (`n_estimators`)	100
Maximum Tree Depth (`max_depth`)	None
Minimum Samples Split (`min_samples_split`)	5
Minimum Samples per Leaf (`min_samples_leaf`)	2

These values represent the configuration that achieved the highest average cross-validation performance during the optimisation process.

4.4.4 Cross-Validation Performance

The optimisation process produced a **best cross-validation accuracy of 96.90%**, indicating excellent predictive performance across the validation folds.

Table 4- 5 Cross-Validation Performance

Metric	Value
Cross-Validation Method	5-Fold Cross-Validation
Optimisation Technique	GridSearchCV
Scoring Metric	Accuracy
Best Cross-Validation Accuracy	**96.90%**

The high cross-validation score suggests that the optimised Random Forest model generalised well during training and was unlikely to have learned patterns specific to a single data partition.

4.4.5 Performance of the Optimised Model

After selecting the optimal hyperparameters, the Random Forest classifier was retrained using the training dataset and subsequently evaluated on the independent testing dataset.

The optimised model achieved a **test accuracy of 94.57%**, representing a slight improvement over the baseline Random Forest model.

Table 4- 6 Performance of the Optimised Random Forest Classifier

METRIC	VALUE
TEST ACCURACY	94.57%

The classification report further demonstrated strong predictive performance across the ten bearing fault classes. Most classes achieved precision, recall and F1-scores close to **1.00**, indicating excellent classification capability.

4.4.6 Discussion of the Optimisation Results

The hyperparameter optimisation process resulted in a modest but measurable improvement in the performance of the Random Forest classifier. The test accuracy increased from **94.35%** for the baseline model to **94.57%** after optimisation, corresponding to an improvement of approximately **0.22 percentage points**.

Although the increase in accuracy was relatively small, it demonstrates that the baseline model had already captured much of the predictive information contained in the statistical vibration features. The optimisation process therefore served to refine the classifier rather than fundamentally alter its behaviour.

The selected hyperparameters contributed to a better balance between model complexity and generalisation. In particular:

- Setting `min_samples_split = 5` reduced the likelihood of creating overly specific branches.
- Using `min_samples_leaf = 2` helped prevent the formation of leaf nodes containing very few samples, thereby improving robustness.
- Retaining `max_depth = None` allowed the trees to grow fully where supported by the data, while the other constraints limited unnecessary complexity.

These findings indicate that careful hyperparameter tuning can produce incremental improvements while maintaining the stability and reliability of the classifier.

4.4.7 Engineering Interpretation

From an engineering perspective, the optimised Random Forest model demonstrates strong potential for vibration-based condition monitoring. Its high classification accuracy and balanced performance across multiple fault classes suggest that it can reliably distinguish between healthy bearings and different fault conditions.

The modest improvement achieved through hyperparameter optimisation also indicates that the statistical vibration features extracted from the CWRU-derived dataset are highly informative. This reinforces the suitability of the selected feature set for predictive maintenance applications in rotating machinery.

4.5 Final Model Performance

4.5.1 Introduction

Following hyperparameter optimisation, the Random Forest classifier was selected as the final predictive model for rolling element bearing fault diagnosis. The optimised model was retrained using the training dataset and evaluated on the independent testing dataset to determine its effectiveness in classifying the ten bearing fault conditions represented in the CWRU bearing dataset.

The objective of this evaluation was to assess the predictive capability of the optimised classifier under previously unseen conditions and determine its suitability for predictive maintenance applications.

4.5.2 Overall Performance of the Final Model

The optimised Random Forest classifier achieved a **test accuracy of 94.57%**, indicating that it correctly classified approximately **95 out of every 100** bearing fault observations in the testing dataset.

Table 4- 7 Overall Performance of the Final Random Forest Model

Performance Metric	Value

Training Samples	1,840
Testing Samples	460
Accuracy	94.57%
Macro Precision	95%
Macro Recall	95%
Macro F1-score	95%

These results demonstrate that the optimised Random Forest classifier achieved a high level of predictive performance while maintaining a balanced trade-off between precision and recall across the ten bearing fault classes.

4.5.3 Interpretation of the Final Model Performance

The overall performance of the optimised Random Forest classifier demonstrates that the selected statistical vibration features provide sufficient discriminatory information for accurate bearing fault classification.

The high accuracy of **94.57%**, combined with balanced macro precision, recall and F1-score values of approximately **95%**, indicates that the classifier achieved both high predictive accuracy and consistent performance across multiple fault categories.

4.5.4 Practical Significance

From an engineering perspective, the developed classifier demonstrates strong potential for use in predictive maintenance systems.

Accurate identification of bearing faults enables maintenance personnel to detect incipient failures before catastrophic damage occurs, reducing equipment downtime, maintenance costs and safety risks. The achieved performance indicates that the proposed machine learning framework can serve as a reliable decision-support tool for vibration-based condition monitoring of rotating machinery.

4.5.5 Model Deployment

The final Random Forest classifier was successfully saved as a serialized model file (`random_forest_model.pkl`), enabling future predictions without the need for retraining.

Saving the model facilitates:

- Rapid deployment in predictive maintenance applications.
- Consistent prediction on new vibration data.
- Reproducibility of experimental results.

- Integration into real-time condition monitoring systems.

This capability represents an important step toward the practical application of machine learning techniques in industrial environments.

4.5.6 Limitations of the Study

Although the developed model achieved strong predictive performance, several limitations should be acknowledged.

Firstly, the study utilised the publicly available CWRU bearing dataset, which was collected under controlled laboratory conditions. Industrial environments often contain additional sources of noise, varying operating conditions and environmental disturbances that may influence model performance.

Secondly, only statistical time-domain vibration features were considered. Incorporating frequency-domain or time-frequency features may further improve classification performance, particularly for fault classes exhibiting similar statistical characteristics.

Finally, the study focused on classical supervised machine learning algorithms. Recent advances in deep learning may offer opportunities for automatic feature extraction and further improvements in classification accuracy when larger datasets are available.

Acknowledging these limitations provides opportunities for future research while placing the findings of this study into their proper context.

4.5.7 Summary

The findings demonstrate that machine learning techniques are highly effective for vibration-based bearing fault diagnosis when combined with carefully selected statistical features. Among the evaluated classifiers, the Random Forest algorithm achieved the strongest overall performance, and hyperparameter optimisation further improved its predictive capability.

Overall, the developed framework satisfies the objectives of the study and demonstrates strong potential for application in intelligent predictive maintenance systems.

CHAPTER FIVE

5 SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

5.1 Summary of the Study

This study was undertaken to develop an accurate and reliable machine learning framework for the diagnosis of rolling element bearing faults using statistical vibration features. Rolling element bearings are critical components of rotating machinery, and their failure can result in significant operational downtime, increased maintenance costs and safety risks. Consequently, the development of intelligent diagnostic systems capable of detecting bearing faults at an early stage is of considerable importance in modern predictive maintenance.

The study began with an extensive review of literature on rolling element bearings, predictive maintenance, vibration analysis, statistical feature extraction and machine learning techniques used in fault diagnosis. The literature review identified a research gap in the comparative evaluation of classical supervised machine learning algorithms using statistical time-domain features for multiclass bearing fault classification.

To address this gap, the publicly available CWRU bearing dataset was selected as the experimental dataset. Exploratory Data Analysis (EDA) confirmed that the dataset contained **2,300 observations**, **nine statistical predictor variables** and **ten bearing fault classes**. The dataset was found to be balanced and free from missing values and duplicate records, making it suitable for supervised learning.

The data preprocessing stage involved label encoding of the target variable, partitioning the dataset into training and testing subsets, and feature standardisation. These steps ensured that the machine learning models were trained and evaluated using a consistent and unbiased framework.

Five supervised machine learning algorithms namely: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours and Support Vector Machine. Were developed and evaluated. Comparative analysis showed that the Random Forest classifier achieved the best overall performance among the baseline models.

To further enhance predictive performance, the Random Forest classifier was optimised using GridSearchCV with five-fold cross-validation. The optimised model achieved a final classification accuracy of **94.57%**, demonstrating strong capability in distinguishing between healthy bearings and multiple fault conditions.

The study therefore demonstrated that statistical vibration features combined with supervised machine learning algorithms can provide an effective and computationally efficient solution for bearing fault diagnosis.

5.2 Summary of Major Findings

The major findings of this study are summarised as follows:

1. **Statistical vibration features were effective for bearing fault diagnosis.**

The statistical time-domain features extracted from the vibration signals provided sufficient information to distinguish between healthy bearings and different fault conditions. The consistently high classification performance achieved by all evaluated machine learning models demonstrates that these features effectively represent the operational state of rolling element bearings.

2. **The selected machine learning algorithms successfully classified bearing fault conditions.**

Five supervised machine learning algorithms: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours and Support Vector Machine, were successfully developed and evaluated. Each classifier achieved a classification accuracy exceeding **90%**, demonstrating that classical supervised machine learning techniques are highly effective for multiclass bearing fault diagnosis when appropriate statistical features are used.

3. **Random Forest achieved the best overall baseline performance.**

Among the evaluated classifiers, the Random Forest algorithm consistently produced the highest performance across the principal evaluation metrics, including accuracy, precision, recall and F1-score. Its ensemble learning approach enabled it to model the complex, non-linear relationships present in the vibration-derived statistical features more effectively than the other classifiers.

4. **Hyperparameter optimisation further improved the performance of the selected model.**

The application of GridSearchCV with five-fold cross-validation identified an improved hyperparameter configuration for the Random Forest classifier. Although the increase in classification accuracy was modest, the optimisation process enhanced the robustness and generalisation capability of the model while maintaining excellent predictive performance on the testing dataset.

5. **The final optimised model demonstrated high predictive capability.**

The optimised Random Forest classifier achieved a final test accuracy of **94.57%**, together with strong precision, recall and F1-score values. The model also demonstrated excellent class-wise performance, correctly identifying most bearing fault categories with a high degree of reliability.

6. **The developed workflow is reproducible and suitable for predictive maintenance applications.**

The study successfully developed a structured machine learning workflow comprising data preprocessing, feature importance analysis, model development, comparative evaluation, hyperparameter optimisation and final model deployment. The modular implementation enhances reproducibility and provides a practical framework that can be adapted to other condition monitoring and fault diagnosis applications involving rotating machinery.

7. **The research objectives were successfully achieved.**

The findings demonstrate that the objectives established at the beginning of the study were fully achieved. The dataset was effectively analysed and prepared, multiple machine learning models were developed and evaluated, the best-performing model was identified and optimised, and a reliable framework for vibration-based bearing fault diagnosis was successfully established.

5.3 Conclusions

Based on the findings of this study, it can be concluded that supervised machine learning techniques provide an effective and reliable approach for the diagnosis of rolling element bearing faults using statistical vibration features. The developed machine learning framework successfully classified multiple bearing fault conditions with a high degree of accuracy, demonstrating the suitability of statistical time-domain features as informative predictors of bearing health.

The comparative evaluation of the five supervised machine learning algorithms showed that all the selected classifiers were capable of performing multiclass bearing fault diagnosis with classification accuracies exceeding 90%. This confirms that classical machine learning algorithms remain valuable tools for condition monitoring and fault diagnosis, particularly when supported by appropriate feature engineering and systematic data preprocessing.

Among the evaluated models, the Random Forest classifier consistently achieved the strongest overall performance. Its ensemble learning architecture enabled it to capture the complex, non-linear relationships present in the vibration-derived statistical features more effectively than the linear, instance-based and single-tree classifiers evaluated in this study. Consequently, Random Forest was identified as the most suitable algorithm for the selected dataset and problem domain.

The application of GridSearchCV for hyperparameter optimisation further enhanced the performance of the selected classifier. Although the improvement in classification accuracy was modest, the optimisation process demonstrated the importance of systematic model tuning in improving generalisation and ensuring robust predictive performance. This finding highlights that model optimisation should be regarded as an essential component of machine learning-based engineering applications rather than an optional refinement.

From an engineering perspective, the developed model demonstrates significant potential for predictive maintenance applications. Accurate classification of bearing fault conditions enables early fault detection, reduces the likelihood of unexpected equipment failure and supports more effective condition-based maintenance strategies. Consequently, the proposed framework contributes to improving equipment reliability, reducing maintenance costs and enhancing operational safety in industrial environments.

Overall, this study successfully achieved its aim and objectives by developing, evaluating and optimising a machine learning-based bearing fault diagnosis system. The findings contribute to the growing body of knowledge on intelligent condition monitoring and demonstrate that carefully engineered statistical vibration features, combined with appropriate supervised learning algorithms can provide accurate, reliable and practical solutions for bearing fault diagnosis.

5.4 Contributions of the Study

This study makes several contributions to the fields of machine learning, vibration-based condition monitoring and predictive maintenance of rotating machinery. These contributions are academic, methodological and practical in nature.

5.4.1 Academic Contributions

This study contributes to the growing body of knowledge on intelligent fault diagnosis by demonstrating the effectiveness of classical supervised machine learning algorithms for multiclass rolling element bearing fault classification using statistical time-domain vibration features.

The findings also reinforce existing research indicating that ensemble learning methods, particularly Random Forest, are well suited to modelling the complex relationships present in vibration-derived statistical features.

5.4.2 Methodological Contributions

One of the major contributions of this study is the development of a structured, modular and reproducible machine learning workflow for bearing fault diagnosis.

The workflow encompasses the complete machine learning pipeline, including:

- Exploratory Data Analysis (EDA)
- Data preprocessing
- Feature importance analysis
- Model development
- Comparative evaluation
- Hyperparameter optimisation
- Final model validation
- Model persistence for deployment

5.4.3 Practical Contributions

From an engineering perspective, the developed machine learning framework provides a practical approach to vibration-based condition monitoring of rolling element bearings.

The high predictive performance achieved by the optimised Random Forest classifier demonstrates that machine learning techniques can effectively support early fault detection, enabling maintenance personnel to identify developing bearing defects before catastrophic failure occurs.

The framework therefore has the potential to contribute to:

- Improved equipment reliability.
- Reduced unplanned machine downtime.
- Lower maintenance and repair costs.
- Enhanced operational safety.
- More effective condition-based maintenance planning.

Although the present study was conducted using a benchmark dataset, the methodology can be adapted to industrial vibration monitoring systems with appropriate data acquisition and validation.

5.4.4 Technical Contributions

This research also contributes a complete implementation framework for supervised machine learning-based bearing fault diagnosis using Python and the Scikit-learn ecosystem.

The implementation includes:

- Statistical feature-based data representation.
- Standardised preprocessing procedures.
- Training and evaluation of multiple classifiers.
- Hyperparameter optimisation using GridSearchCV.
- Performance assessment using accuracy, precision, recall, F1-score, confusion matrices and classification reports.
- Model serialisation for future deployment.

This end-to-end implementation demonstrates a complete engineering workflow, bridging theoretical concepts with practical application.

5.4.5 Educational Contribution

Beyond its technical outcomes, this study provides a comprehensive example of how machine learning can be applied to a traditional mechanical engineering problem.

The documented workflow, implementation strategy and technical report may serve as a useful reference for undergraduate and postgraduate students interested in predictive maintenance, machine learning and intelligent manufacturing. By illustrating each stage of the development process, the study offers a practical learning resource that can support future educational and research activities.

5.4.6 Overall Contribution

In summary, this study demonstrates that statistical vibration features combined with supervised machine learning algorithms provide a reliable, efficient and reproducible approach to rolling element bearing fault diagnosis.

The research contributes not only an effective predictive model but also a structured methodology that integrates engineering knowledge with data-driven techniques. This integration supports the growing adoption of intelligent maintenance systems within the context of modern Industry 4.0 manufacturing and condition monitoring practices.

5.5 Recommendations

Based on the findings and conclusions of this study, the following recommendations are proposed for industry, researchers and engineering practitioners.

5.5.1 Recommendations for Industrial Practice

Industries that rely on rotating machinery should consider integrating machine learning-based fault diagnosis systems into their predictive maintenance programmes. The developed Random Forest classifier demonstrated strong performance in identifying bearing fault conditions and has the potential to improve maintenance decision-making by providing early warnings of developing faults.

The adoption of intelligent fault diagnosis systems can help industries reduce unplanned equipment downtime, minimise maintenance costs, improve equipment reliability and enhance operational safety.

5.5.2 Recommendations for Condition Monitoring Systems

Organisations implementing condition monitoring programmes should complement traditional vibration analysis with machine learning techniques. Rather than relying solely on manual interpretation of vibration signals, automated diagnostic models can assist maintenance engineers by rapidly identifying bearing health conditions and supporting timely maintenance interventions.

Combining machine learning predictions with expert engineering judgement can further improve the reliability of maintenance decisions.

5.5.3 Recommendations for Engineering Practice

Mechanical engineers involved in machinery maintenance and reliability engineering should consider incorporating data-driven diagnostic techniques into conventional maintenance strategies. The integration of vibration analysis, statistical feature extraction and supervised machine learning can enhance fault detection accuracy while supporting the transition from preventive maintenance to condition-based and predictive maintenance approaches.

Engineering organisations should also invest in training personnel in data analytics and machine learning to facilitate the adoption of intelligent maintenance technologies.

5.5.4 Recommendations for Researchers

Researchers working in machine learning-based fault diagnosis should evaluate multiple machine learning algorithms before selecting a final predictive model. The comparative analysis conducted in this study demonstrated that algorithm selection has a significant influence on diagnostic performance.

Researchers are also encouraged to perform systematic hyperparameter optimisation rather than relying solely on default model settings, as appropriate tuning can improve model robustness and predictive capability.

5.5.5 Recommendations for Educational Institutions

Universities and engineering training institutions should strengthen the integration of artificial intelligence and machine learning into mechanical engineering curricula. As intelligent maintenance systems become increasingly important in modern industry, graduates should be equipped with practical skills in data preprocessing, machine learning model development and predictive maintenance technologies.

Project-based learning that combines mechanical engineering principles with data science techniques can help prepare students for the demands of Industry 4.0 and smart manufacturing.

5.5.6 General Recommendation

The machine learning framework developed in this study should be regarded as a decision-support tool rather than a replacement for engineering expertise. Machine learning models provide valuable diagnostic insights, but maintenance decisions should continue to consider operational conditions, equipment history and professional engineering judgement.

Future implementation of such systems should therefore combine intelligent algorithms with human expertise to achieve safe, reliable and effective predictive maintenance.

5.6 Suggestions for Future Research

Although this study successfully demonstrated the effectiveness of supervised machine learning techniques for rolling element bearing fault diagnosis using statistical vibration features, several opportunities exist for further investigation.

5.6.1 Incorporation of Frequency-Domain and Time-Frequency Features

This study utilised statistical time-domain vibration features for fault classification. Future research should investigate the integration of frequency-domain and time-frequency feature extraction techniques, such as the Fast Fourier Transform (FFT), Wavelet Transform and Empirical Mode Decomposition (EMD). These approaches may capture additional characteristics of bearing vibration signals and improve the discrimination of fault classes with similar statistical properties.

5.6.2 Evaluation of Deep Learning Techniques

The present study focused on classical supervised machine learning algorithms. Future studies could evaluate deep learning architectures, including Artificial Neural Networks (ANNs), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks. Such models may automatically learn complex feature representations from vibration data and could provide improved performance, particularly for large-scale datasets.

5.6.3 Validation Using Real Industrial Datasets

The experiments conducted in this study were based on the publicly available Case Western Reserve University (CWRU) bearing dataset, which was collected under controlled laboratory conditions. Future research should validate the developed methodology using vibration data collected from operational industrial machinery under varying loads, speeds and environmental conditions. Such validation would provide a stronger assessment of the model's robustness and practical applicability.

5.6.4 Development of Real-Time Fault Diagnosis Systems

Future work should investigate the implementation of the developed machine learning framework within real-time condition monitoring systems. Integrating vibration sensors, data acquisition hardware and machine learning models into an online monitoring platform would enable continuous equipment health assessment and support timely maintenance decision-making in industrial environments.

5.6.5 Investigation of Additional Machine Learning Algorithms

Although this study evaluated five widely used supervised machine learning algorithms, future research may explore additional techniques such as Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), CatBoost and ensemble stacking methods. Comparing these algorithms with the Random Forest classifier may provide further insights into their suitability for bearing fault diagnosis.

5.6.6 Automated Feature Selection and Engineering

Future studies should investigate advanced feature selection and feature engineering techniques to identify the most informative vibration characteristics while reducing computational complexity. Methods such as Recursive Feature Elimination (RFE), mutual information analysis and embedded feature selection techniques could improve model efficiency without compromising predictive performance.

5.6.7 Deployment in Industry 4.0 Environments

Future research should examine the integration of machine learning-based fault diagnosis systems into Industry 4.0 and Industrial Internet of Things (IIoT) environments. Combining intelligent diagnostic models with cloud computing, edge computing and wireless sensor networks may facilitate scalable, remote and continuous monitoring of industrial machinery across multiple locations.

5.6.8 Explainable Artificial Intelligence (XAI)

As machine learning models become increasingly integrated into industrial maintenance systems, understanding how models arrive at their predictions is becoming more important. Future research should investigate Explainable Artificial Intelligence (XAI) techniques, such as SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations), to improve transparency, interpretability and trust in machine learning-based fault diagnosis systems.

5.6.9 Final Remark

The findings of this study demonstrate that machine learning provides a promising approach to vibration-based bearing fault diagnosis. Continued research that incorporates more diverse datasets, advanced feature extraction techniques, modern learning algorithms and real-world deployment strategies will further enhance the reliability, scalability and industrial applicability of intelligent predictive maintenance systems.

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APPENDICES

APPENDIX A

CASE WESTERN RESERVE UNIVERSITY (CWRU) BEARING DATASET

A.1 Description of the Dataset

The dataset used in this study was obtained from the **Case Western Reserve University (CWRU) Bearing Data Center**, a widely recognised benchmark dataset for research on bearing fault diagnosis and predictive maintenance. The dataset was generated using a controlled laboratory test rig equipped with an electric motor, torque transducer, dynamometer and rolling element bearings.

To simulate different bearing health conditions, faults of varying severities were introduced into the bearings using electrical discharge machining (EDM). Vibration signals were then collected under different operating conditions using accelerometers mounted on the drive-end and fan-end bearing housings. These signals represent both healthy and faulty bearing conditions and have been extensively used in machine learning and condition monitoring research.

A.2 Dataset Characteristics

Table A.1: Summary of the Dataset

Characteristic	Description
Dataset Source	Case Western Reserve University Bearing Data Center
Application	Bearing Fault Diagnosis
Data Type	Statistical vibration features
Number of Samples	2,300
Number of Predictor Variables	9
Number of Target Classes	10
Classification Type	Multiclass Classification

A.3 Description of Variables

Variable	Description
Max/Min	Peak vibration amplitudes. Faults often cause spikes
Mean	Average value of the vibration signal
Standard Deviation	Measure of signal dispersion

RMS	Root Mean Square value of the vibration signal
Skewness	Measure of signal asymmetry
Kurtosis	Measure of signal peakedness
Crest Factor	Ratio of peak value to RMS
Form	Shape factor = RMS/mean_rectified. Related to signal shape
Fault	Bearing condition (Target Variable)

A.4 Sample of the Dataset

Table A.2: Sample Records from the Processed Bearing Dataset

max	min	mean	sd	rms	skewness	kurtosis	crest	form	fault
0.35986	-0.4189	0.01784	0.122746	0.124006	-0.11857	-0.04222	2.901946	6.950855	Ball_007_1
0.46772	-0.36111	0.022255	0.132488	0.134312	0.174699	-0.08155	3.482334	6.035202	Ball_007_1
0.46855	-0.43809	0.02047	0.149651	0.151008	0.040339	-0.27407	3.102819	7.376926	Ball_007_1
0.58475	-0.54303	0.02096	0.157067	0.158422	-0.02327	0.134692	3.691097	7.558387	Ball_007_1
0.44685	-0.57891	0.022167	0.138189	0.139922	-0.08153	0.402783	3.193561	6.312085	Ball_007_1
0.43726	-0.44435	0.021119	0.138763	0.140328	-0.13133	-0.16856	3.11599	6.644538	Ball_007_1
0.45353	-0.49129	0.021464	0.138461	0.140082	-0.11417	0.308107	3.237609	6.526352	Ball_007_1
0.43955	-0.45228	0.02086	0.15012	0.151526	-0.02196	-0.2723	2.90082	7.263885	Ball_007_1
0.49233	-0.37217	0.020244	0.145361	0.146729	0.074174	-0.42181	3.355377	7.248013	Ball_007_1
0.37154	-0.49087	0.018105	0.136393	0.137556	-0.13624	-0.09789	2.701005	7.597902	Ball_007_1

A.5 Source of the Dataset

Source: The dataset used in this study was obtained from the Case Western Reserve University Bearing Data Center and processed by the researcher to extract statistical vibration features for machine learning-based bearing fault classification.

APPENDIX B
COMPLETE PYTHON CODE

The complete Python code can be accessed via the GitHub Repository link below:

https://github.com/McImaga/bearing-failure-prediction